

卒業論文

Fundamental Study on PSF
Characterization and AI-based Image
Enhancement for Aerial Imaging
Optical Systems

(空中結像光学系の PSF 特性評価と AI による画質向上処理
に関する基礎的研究)

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Ahmad Aiman Mirza bin Ahmad Zamri

宇都宮大学工学部

基盤工学科 情報電子オプティクスコース

情報学科分野

内容梗概

本研究では、空中表示技術 AIRR (Aerial Imaging by Retro-Reflection) における PSF (Point Spread Function) の特性評価と、その特性を考慮した AI を用いた画像強調手法に関する基礎的検討について報告する。空中表示技術 AIRR は空中に実像を形成でき、複数人が同時に観察可能であるという特徴を有する一方、回折や光学系の影響により像のボケが生じるため、画質の向上が重要な課題となっている。このような解像度劣化を定量的に扱うためには、AIRR の PSF 特性を適切に把握することが不可欠である。

AIRR により形成される空中像は物理的なスクリーンを持たないため、PSF の直接計測が困難であり、カメラとリレー光学系を用いた間接的な測定手法が用いられる。しかし、この測定過程において取得される PSF には、センサのオフセットや背景光などに起因する、空間的にほぼ一様な基線成分（以下、ペDESTAL成分）が重畳されることがある。本研究では、測定 PSF に含まれるペDESTAL成分に着目し、その発生要因を整理するとともに、PSF 特性評価に与える影響について検討した。

具体的には、測定 PSF に対してペDESTAL成分を加算的な一定オフセットとしてモデル化し、前処理としてペDESTAL成分除去を適用した。除去前後の PSF を比較した結果、ペDESTAL成分が PSF の基線に影響を与え、物理的な PSF 解釈や数値的なぼけシミュレーションを困難にする要因となることを明らかにした。これらの結果から、測定過程に起因するペDESTAL成分は、PSF の本質的特性とは異なる非物理的な成分、すなわちペDESTALアーティファクトとして扱う必要があることを示した。

さらに、仮定した PSF 特性が AI を用いた画像強調に与える影響について検討した。AIRR における一般的なぼけ傾向を想定した合成 PSF を用いて劣化画像を生成し、それらを用いて畳み込みニューラルネットワークによる画像強調の学習および評価を行った。その結果、このようなぼけ仮定の下において AI を用いた事前復元が可能であること、および仮定する PSF 特性の違いが復元挙動に差異を生じさせることを確認した。これらの結果から、PSF 特性の仮定が AI を用いた画像強調において重要な役割を果たすことが示唆され、AIRR システムに対する AI を用いた画像品質改善手法を検討する際には、PSF 特性を慎重に考慮する必要があることが示された。

空中結像光学系の PSF 特性評価と AI による 画質向上処理に関する基礎的研究

ABSTRACT

This study reports a fundamental investigation of PSF (Point Spread Function) characterization and AI-based image enhancement for AIRR (Aerial Imaging by Retro-Reflection). AIRR is an aerial display technology capable of forming images in free space that can be simultaneously observed by multiple viewers. However, image blur caused by diffraction and optical system characteristics degrades image quality in practical AIRR systems, and accurate PSF characterization is essential for understanding image degradation and informing the design of AI-based image enhancement.

Since AIRR images are formed without a physical screen, direct measurement of the PSF is difficult, and indirect measurement using relay optics and an image sensor is commonly employed. In such measurements, the acquired PSF often contains a spatially uniform baseline intensity component originating from the measurement chain. In this study, this component is referred to as a pedestal component, and its influence on PSF characterization is investigated.

Furthermore, the influence of assumed PSF characteristics on AI-based image enhancement is examined using synthetic blur kernels and a convolutional neural network. The results demonstrate that AI-based pre-restoration is feasible and that restoration behavior depends on PSF assumptions, highlighting the importance of careful PSF consideration.

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1 Introduction

1.1 Research Background

In recent years, touchless interaction technologies have attracted increasing attention in public and industrial environments where hygiene, safety, and ease of use are critical. Public interfaces such as retail terminals, elevators, and information kiosks are operated by a large and unspecified number of users, making conventional contact-based interfaces undesirable from both hygienic and maintenance perspectives. In response to these demands, non-contact user interfaces have been actively explored and introduced into real-world systems, including point-of-sale terminals, elevator control panels, and public information displays [1,2,3]. These trends have motivated the development of display technologies that enable users to visually recognize and interact with information without direct physical contact.

Among various approaches to realizing touchless interfaces, aerial display technologies—which present visual information in free space without a physical screen—have been widely investigated. By forming images that appear to float in mid-air, aerial displays allow users to perceive information in an intuitive and spatially accessible manner. A number of optical methods for aerial image formation have been proposed, including systems based on dihedral corner reflector arrays [4], face-to-face mirror configurations [5], and micro-reflector plate structures [6]. Although these approaches differ in optical principle and system configuration, they share the common objective of presenting visually recognizable images in free space for intuitive interaction.

Aerial Imaging by Retro-Reflection (AIRR) is one representative aerial display technology that forms real images in mid-air using a retro-reflective optical configuration [7]. AIRR enables wide viewing angles and simultaneous observation by multiple viewers, making it suitable for applications such as signage, interpersonal displays, and interactive systems. Based on these characteristics, various AIRR-based systems have been demonstrated, including aerial LED signage [8], both-sided aerial displays [9], and interactive communication platforms [10]. These studies indicate that AIRR is a practical and versatile aerial display technology for touchless user interfaces.

Despite these advantages, image quality remains a critical issue in aerial display systems, including AIRR. In practical applications, insufficient image sharpness and reduced contrast degrade visibility and limit usability, thereby restricting the effectiveness of aerial displays as

user interfaces. As a result, improving image quality is an essential requirement for the wider adoption and deployment of aerial display technologies.

From an optical perspective, the image formed by an optical system is not a perfect reproduction of the original object but is inevitably degraded by physical factors inherent to the system [11]. Among these factors, spatial blur caused by diffraction and system aberrations constitutes a fundamental limitation that cannot be completely eliminated through improvements in device fabrication alone [12]. Such image degradation is commonly characterized by the point spread function (PSF), which represents the impulse response of the optical system and describes the relationship between the original image and the observed image through convolution [11,13].

From a signal processing standpoint, image blur can be modeled as a linear shift-invariant degradation under appropriate conditions, enabling PSF-based analysis and restoration techniques [14]. Based on this model, inverse filtering and deconvolution methods have been extensively studied to compensate for system-induced blur in optical imaging and display systems [15,16]. In the context of AIRR, PSF-based deconvolution has been shown to improve image sharpness when suitable system characterization is available [16].

However, accurate PSF characterization in AIRR systems presents unique challenges. Because AIRR images are formed in free space without a physical image plane, PSFs are typically measured indirectly using relay optics and image sensors. During this measurement process, background components such as sensor offset and stray illumination can be superimposed on the measured PSF, resulting in a spatially uniform baseline component that distorts PSF characteristics and hinders physically consistent interpretation [17].

In parallel with analytical restoration approaches, recent advances in data-driven image processing, particularly convolutional neural networks (CNNs), have enabled image restoration methods that do not rely solely on explicit analytical inversion [18,19]. These approaches offer the potential to compensate for complex and non-ideal blur characteristics when trained on degraded image data generated using assumed or measured PSFs. Nevertheless, the effectiveness of such AI-based methods fundamentally depends on the physical validity and consistency of the PSF models used during training and evaluation.

Therefore, a systematic understanding of PSF characteristics and their influence on both analytical and AI-based image enhancement methods is essential for developing reliable

image quality improvement techniques for aerial imaging systems such as AIRR.

1.2 Research Objective

The objective of this study is to conduct a fundamental investigation into the relationship between point spread function (PSF) characterization and AI-based image enhancement in aerial imaging optical systems, with particular focus on AIRR systems. This study examines the feasibility of AI-based pre-restoration for AIRR image formation using simplified and synthetic PSF models, building on prior AIRR deblurring and computational blur compensation approaches [16]. In parallel, the study investigates the physical validity of experimentally acquired PSFs in AIRR systems, focusing on non-physical pedestal components originating from sensor offset, background illumination, and measurement-side artifacts [17]. A practical preprocessing framework is proposed to remove these components while preserving the intrinsic PSF structure, enabling physically meaningful use of measured PSFs in image simulation and restoration. Through these investigations, this thesis clarifies the conditions under which PSFs can be reliably used for computational image analysis and establishes a physically consistent foundation for future PSF-aware AI-based image enhancement based on both analytical restoration theory and data-driven image restoration frameworks [15,16,18].

1.3 Thesis Structure

The structure of this thesis is outlined as follows:

Chapter 1 presents the research background and objectives of this study, focusing on image quality degradation in aerial imaging optical systems and the role of point spread function (PSF) characterization and AI-based image enhancement in AIRR systems.

Chapter 2 describes the fundamental principles of Aerial Imaging by Retro-Reflection (AIRR) and discusses the origin of image blur in AIRR systems. The concept of the PSF as a descriptor of blur is introduced to provide the physical basis for subsequent analysis.

Chapter 3 focuses on PSF measurement in AIRR systems and investigates pedestal artifacts arising from the measurement process. The origin and characteristics of these non-physical components are analyzed, and a practical pedestal subtraction method for physically consistent PSF preprocessing is presented and discussed.

Chapter 4 examines the feasibility of AI-based pre-restoration for AIRR image blur using synthetic PSF models. A convolutional neural network framework is introduced, and experimental results using simplified and physically derived synthetic PSFs are presented to evaluate the effectiveness of pre-restoration under assumed blur conditions.

Chapter 5 concludes the thesis by summarizing the main findings of this study and outlining directions for future work related to PSF characterization and AI-based image enhancement for aerial imaging optical systems.

2 Aerial Imaging by Retro-Reflection (AIRR)

2.1. Introduction

This chapter describes the fundamental structure and optical principles of Aerial Imaging by Retro-Reflection (AIRR). AIRR is an aerial display technology that forms real images in free space by utilizing retro-reflectors and a beam splitter [7].

AIRR has been proposed as a practical aerial display method, and its feasibility has been demonstrated through various systems and applications, including signage and interactive displays [8,9,10].

This chapter first explains the characteristics and structure of retro-reflective elements used in AIRR systems, followed by the basic optical configuration and image formation principle of AIRR. In addition, optical see-through AIRR, and polarization-modulated AIRR are introduced [19,20]. Furthermore, the physical origins of image blur and spatial resolution degradation in AIRR systems are also discussed [16].

2.2. Retro-Reflector Characteristics and Structure

A retro-reflector is a key optical component in AIRR systems. Unlike a conventional mirror, which reflects incident light according to the law of specular reflection, a retro-reflector returns incident light toward its original source direction [6,7]. Figure 1 illustrates the difference between ordinary reflection and retro-reflection. In the case of a mirror, the reflected ray forms an angle equal to the incident angle with respect to the surface normal, as shown in Fig. 1(a). In contrast, a retro-reflector reflects light such that the outgoing ray propagates approximately along the incident path, regardless of the angle of incidence, as shown in Fig. 1(b).

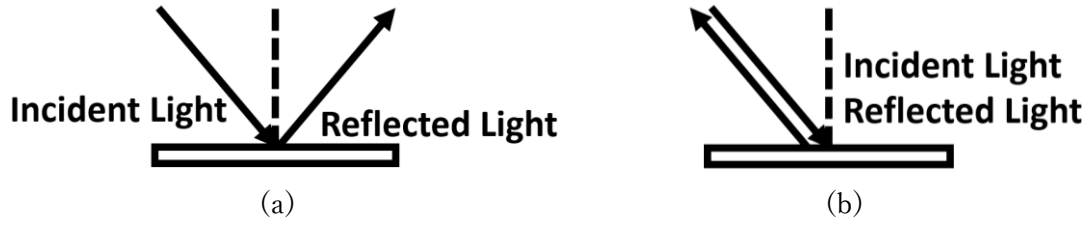


Fig. 1 (a) Normal reflection and (b) retro-reflection.

Retro-reflectors are broadly classified into two structural types: bead-type and microprism-type retro-reflectors, as shown in Fig. 2(a) and Fig. 2(b) [6,9].

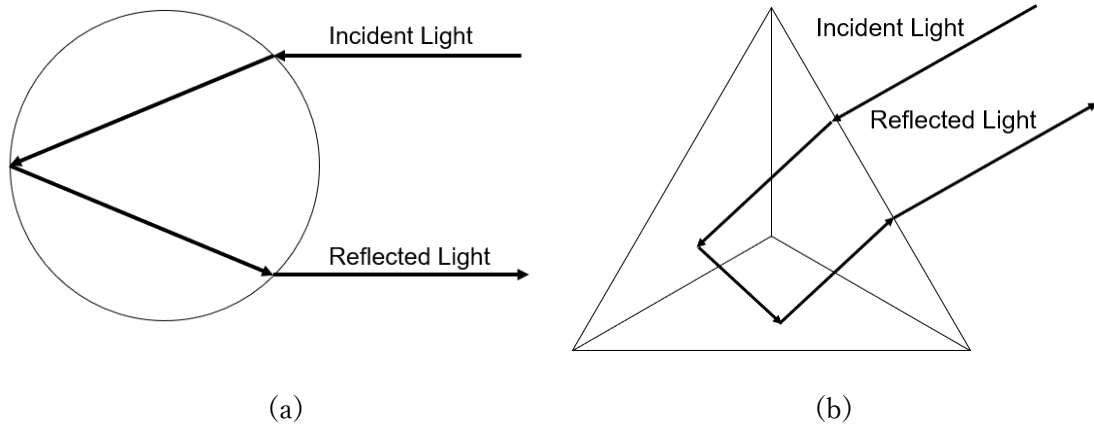


Fig. 2(a) Beads-Type Retro-Reflector and (b) Prism-Type Retro-Reflector

Bead-type retro-reflectors consist of arrays of microscopic glass beads with a high refractive index. Incident light is refracted at the bead surface, reflected by a backing reflective layer, and refracted again upon exiting the bead, resulting in retro-reflection approximately parallel to the incident direction [6].

Microprism-type retro-reflectors, on the other hand, are composed of arrays of corner-cube or triangular prism structures. Incident light entering a microprism undergoes refraction at the prism surface, followed by multiple internal reflections within the prism, and is finally refracted again upon exit. This sequence of refraction and internal reflection causes the light to return toward the incident direction with high directional control [6,9]. A microscopic zoom in of a typical microprism-type retro-reflector is shown in Fig 3.

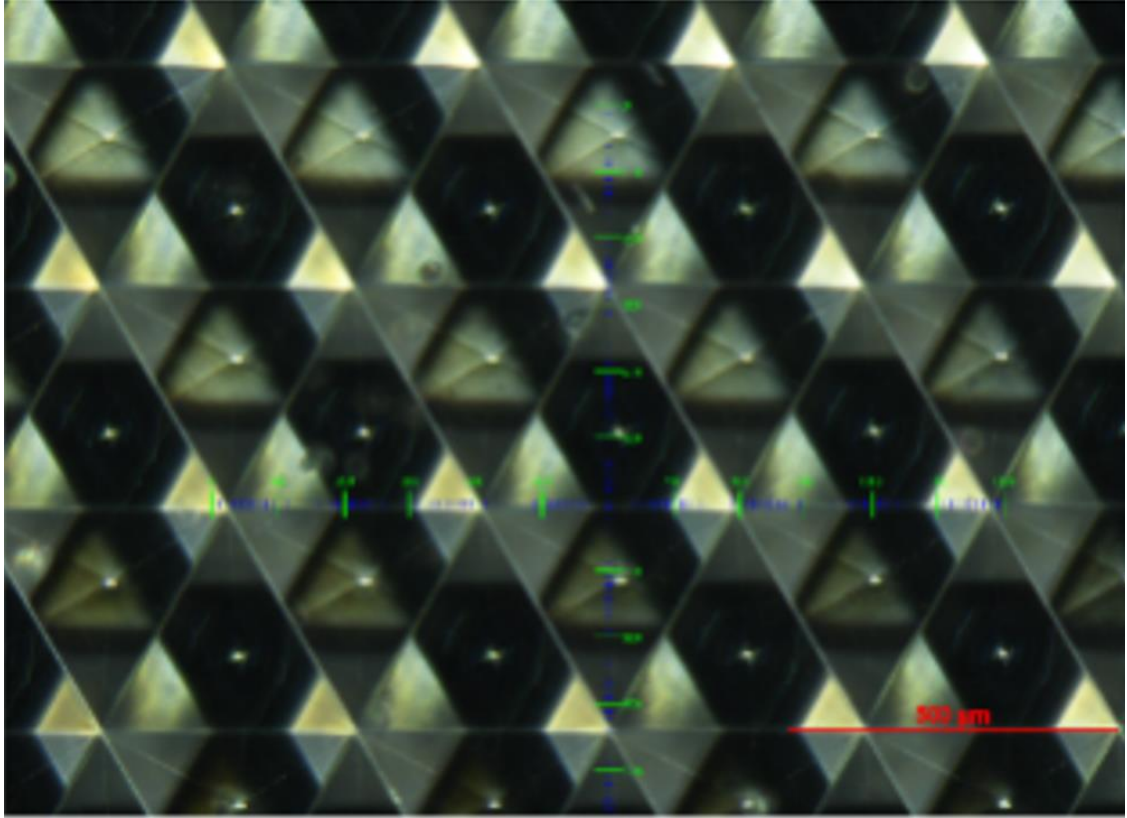


Fig. 3 Microscopic Zoom in of Prism-Type Retro-Reflector

In general applications such as traffic signage, retro-reflectors are designed to provide a certain angular spread to ensure visibility from various directions. In contrast, AIRR systems require retro-reflectors with stronger directional characteristics to enable precise image reconstruction in midair [6,7]. Therefore, this study employs microprism-type retro-reflectors, which provide higher directional retro-reflection and are suitable for aerial image formation in AIRR systems [6,7].

2.3. AIRR Structure and Principle

Aerial Imaging by Retro-Reflection (AIRR) is an aerial image formation technique that exploits the retro-reflective property of optical elements to reconstruct images in free space [8,9,10]. A typical AIRR optical system consists of three main components: a light source, a beam splitter, and a retro-reflector. The basic optical configuration and image formation principle are illustrated in Fig. 4 [9,10].

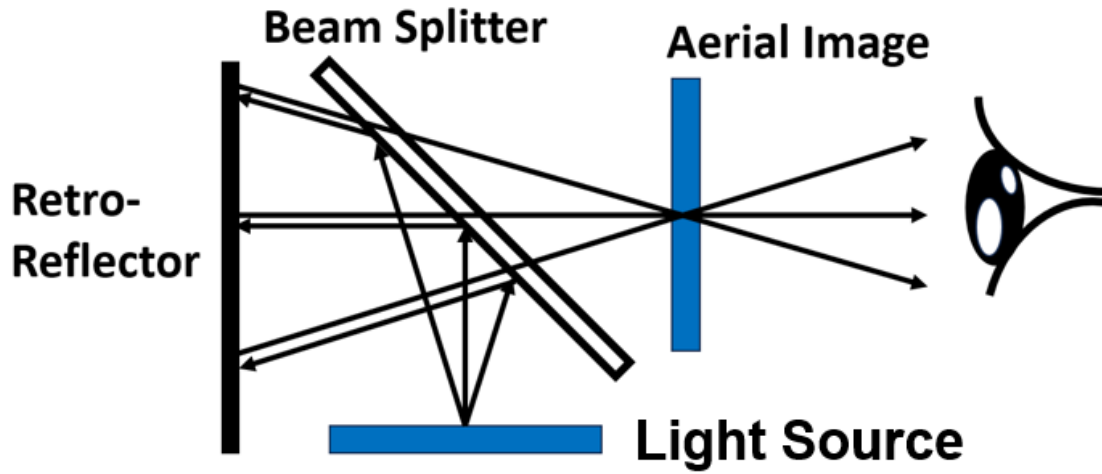


Fig. 4 Principle of AIRR

Light emitted from the light source first propagates toward the beam splitter, where it is divided into reflected and transmitted components. The reflected light is directed toward the retro-reflector, which returns the incident light back along its original propagation direction due to the retro-reflection property described in Section 2.2. The retro-reflected light then returns to the beam splitter and is again divided into reflected and transmitted components [9].

The transmitted light emerging from the beam splitter converges at a position that is plane-symmetric to the original light source with respect to the beam splitter surface, forming a real aerial image in free space [8,10]. As a result, the observer perceives an image that appears to float in midair without the presence of a physical screen at the image plane.

This optical configuration enables aerial image formation using a simple arrangement of optical components and provides flexibility in system layout and scalability [8]. However, because the light passes through multiple optical elements and is subject to finite aperture effects and diffraction, the formed aerial image inevitably exhibits optical degradation, which is discussed in subsequent sections [15,16].

2.4. See Through AIRR

In a see-through AIRR optical system, the light emitted from the light source first transmits through the beam splitter and then reaches the retro-reflector, as shown in Fig. 5. Since the incident light passes through the beam splitter before retro-reflection, the rear side

of the optical system remains optically transparent, allowing objects located behind the system to be directly observed through the optical path. Figure 6 shows a sample see through AIRR apparatus.

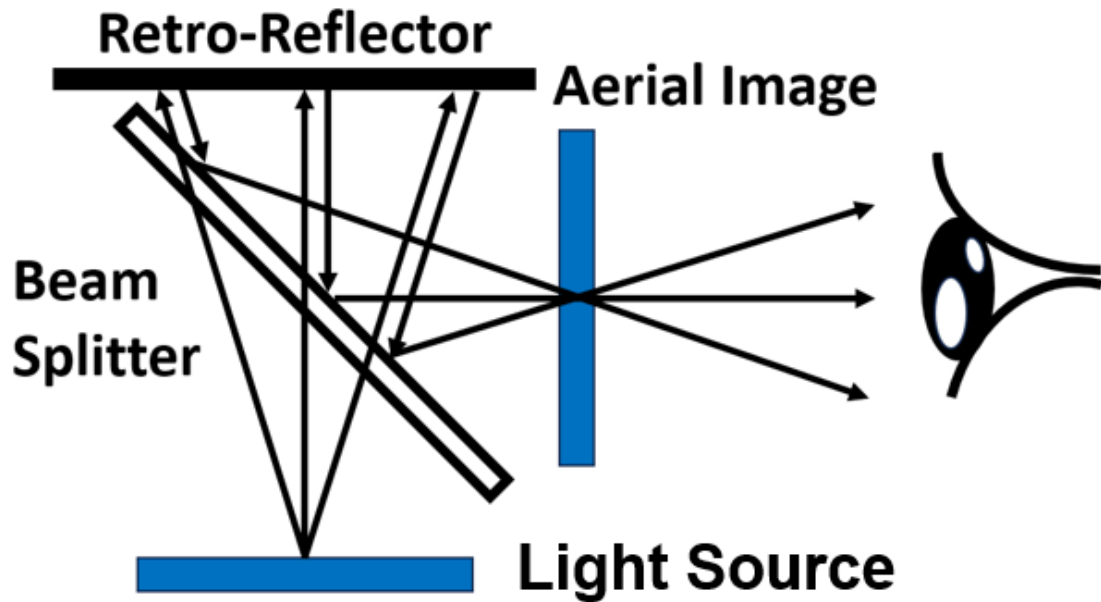


Fig. 5 Principle of See-Through AIRR

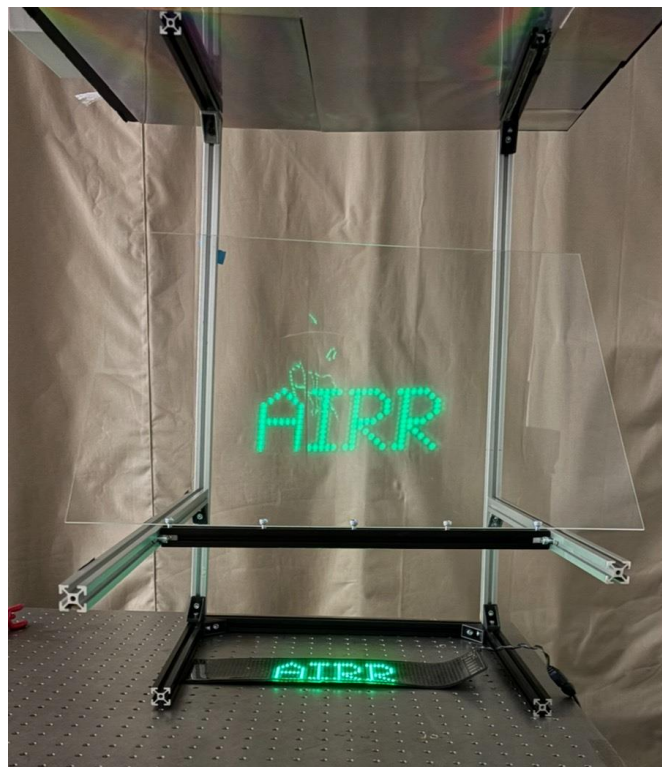


Fig. 6 See-Through AIRR Apparatus

After being reflected by the retro-reflector, the light returns toward the beam splitter due to the retro-reflection property described in Section 2.2. At the beam splitter, the returned light is reflected and converges to form an aerial image at a plane-symmetric position with respect to the original light source, as shown in Fig. 5. As a result, an aerial image can be formed in free space while maintaining optical see-through capability of the system.

This see-through AIRR configuration enables flexible system installation and has been reported as a practical approach for applications requiring both aerial image presentation and background visibility, such as interactive displays and public-space interfaces [19].

2.5. AIRR using Polarizers

In a conventional AIRR optical system, the use of a beam splitter inherently reduces light efficiency because only a portion of the incident light contributes to aerial image formation. As a result, the brightness of the formed aerial image is significantly lower than that of the original light source [7].

To address this limitation, polarization-modulated AIRR systems have been proposed. In these systems, polarization control elements such as linear polarizers and wave plates are introduced into the optical path to selectively route light based on its polarization state. By appropriately designing the polarization configuration, light loss at the beam splitter can be reduced, thereby improving the brightness of the aerial image [20].

In a typical polarization-modulated AIRR configuration, the polarization state of the incident light is controlled such that light traveling toward the retro-reflector and light returning toward the observer experience different polarization-dependent transmission and reflection characteristics at the beam splitter. This approach enables more efficient utilization of the light emitted from the source while preserving the plane-symmetric image formation principle of AIRR [20].

2.6. Image Degradation in AIRR Systems

Although AIRR enables the formation of aerial images in free space, the quality of the formed image is inherently degraded compared with that of the original source image. This degradation appears primarily as optical blur and a loss of spatial resolution and arises

from fundamental physical properties of the AIRR optical system rather than from imperfections in alignment or implementation [11,12].

One major source of image degradation in AIRR systems is diffraction caused by finite apertures in the optical path. Light propagating through the beam splitter and retro-reflective elements experiences diffraction, which leads to spatial spreading of the optical field. As a result, even an ideal point source does not form a perfectly localized point in the aerial image plane but instead produces a spatially extended intensity distribution, imposing a fundamental limit on resolution [11,12].

In addition to diffraction, geometrical ray shifts within the microprism array of the retro-reflective element contribute significantly to image blur. Microprism-type retro-reflectors consist of discrete prism units with finite aperture size. When light enters different regions of a microprism, variations in optical path length and reflection geometry cause lateral displacement of the returned rays, as shown in Fig. 7 [6,21]. When integrated over the entire retro-reflective surface, these geometrical shifts lead to effective aberrations and spatial spreading of the reconstructed image.

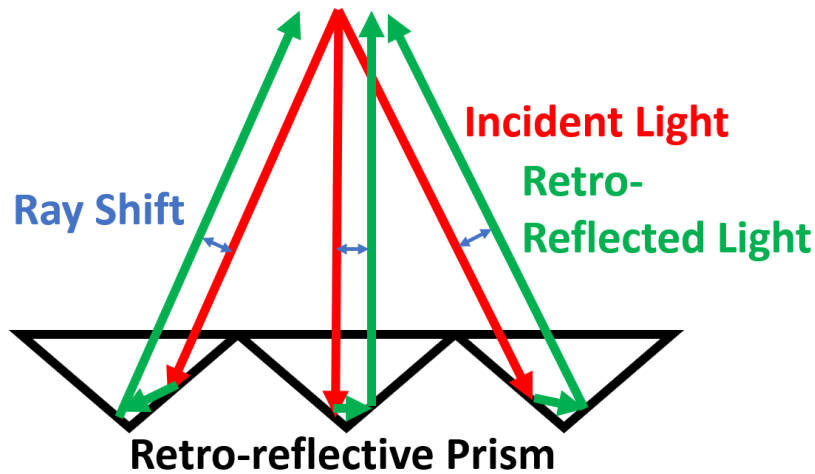


Fig. 7 Geometrical Ray-Shift

Imperfections in the retro-reflective microstructures further exacerbate image degradation. In practical devices, manufacturing tolerances, surface roughness, and deviations from ideal prism geometry introduce non-uniform reflection and scattering. These effects reduce the directionality of the retro-reflected light, resulting in additional loss of contrast and increased blur in the aerial image [6].

The spatial resolution of AIRR displays is also governed by a trade-off related to the aperture size of the micropisms. When the prism aperture size is large, geometrical ray shifts become more pronounced, increasing aberration-induced blur. Conversely, when the prism aperture size is reduced, diffraction effects become stronger, leading to increased diffraction-induced spreading [11,21]. Because these effects exhibit a trade-off relationship, improvements in resolution through retro-reflector design alone are fundamentally limited.

Taken together, these effects cause an AIRR system to respond to a point input not with an ideal point image, but with a spatially extended response that reflects the combined influence of diffraction, geometrical effects, and microstructural imperfections. This response is naturally described by the point spread function (PSF), which provides a physically meaningful basis for analyzing image blur, evaluating resolution limits, and performing numerical image simulation [12,13,14].

2.7. Summary

This chapter introduced the fundamental principles of Aerial Imaging by Retro-Reflection (AIRR) and the optical mechanisms underlying aerial image formation. First, the characteristics and structure of retro-reflectors were described, followed by an explanation of the optical configuration and image formation principle of AIRR systems. Representative variations of AIRR systems were also briefly introduced to provide contextual background.

Furthermore, the origins of image degradation in AIRR systems were discussed, with emphasis on diffraction due to finite apertures, geometrical ray shifts within micropism arrays, and imperfections in retro-reflective microstructures. The trade-off between diffraction effects and geometrical aberrations inherent to retro-reflector design was highlighted, demonstrating fundamental limitations on the achievable spatial resolution of AIRR systems.

3 AI-Based Pre-restoration using Synthetic PSF

3.1 Introduction

Previous studies on image quality improvement in Aerial Imaging by Retro-Reflection (AIRR) systems have demonstrated that image blur caused by diffraction in the optical system can be effectively mitigated through pre-restoration based on analytically derived inverse filters. In particular, Kikuta et al. proposed a pre-correction framework in which blur characteristics quantified by the point spread function (PSF) are compensated

using Wiener deconvolution applied to the input image prior to aerial imaging .

While Wiener deconvolution provides a physically interpretable solution when the PSF is accurately known, it requires explicit PSF estimation, frequency-domain processing, and careful parameter tuning. These requirements limit its flexibility when blur characteristics vary or when PSF estimation becomes unstable. As an alternative approach, data-driven methods based on deep learning have the potential to implicitly learn inverse blur mappings without explicitly constructing analytical inverse filters.

The objective of this chapter is to investigate the feasibility of replacing analytical Wiener-based pre-restoration with an AI-based pre-restoration framework. Rather than directly addressing experimentally measured PSFs, this chapter focuses on controlled numerical experiments using synthetic PSFs with known physical validity. By restricting the evaluation to analytically defined blur models, the capability of AI to learn PSF-dependent pre-restoration can be examined without the confounding effects of measurement noise or experimental artifacts.

In this study, multiple synthetic blur models, including Gaussian blur and diffraction-based PSFs derived from Fraunhofer theory, are used to generate degraded images through convolution. A convolutional neural network is then trained to perform pre-restoration such that the output image, when blurred by the known PSF, approximates the original target image. The performance of the AI-based pre-restoration is evaluated and compared qualitatively and quantitatively against conventional restoration behavior.

The results presented in this chapter serve as a proof of feasibility for AI-based pre-restoration under idealized PSF conditions. By demonstrating that AI can learn effective inverse mappings for physically valid blur models.

3.2 Experiment

3.2.1. Gaussian Blur

As a baseline blur model, a 2D isotropic Gaussian PSF is employed due to its simplicity and its ability to represent generic low-pass optical blur [14]. Gaussian PSFs are widely used in image processing and optical system modeling as a convenient approximation of spatially invariant blur. The continuous Gaussian PSF $K_G(x, y, \sigma)$ is defined as

$$K_G(x, y, \sigma) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2+y^2}{2\sigma^2}\right), \quad (1)$$

where σ denotes the standard deviation controlling the blur strength. The Gaussian PSF is visually shown in Fig. 8 as its logarithmic scale using `cmap=veridis`.

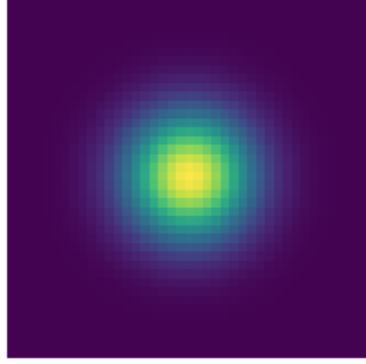


Fig. 8 Gaussian PSF

In numerical implementation, K_G is sampled on a finite odd-sized grid and normalized to enforce unit total energy to satisfy the energy conservation property of a physically valid PSF [12], as shown in the following equation,

$$K_{G,\text{norm}}[m, n] = \frac{K_G[m, n]}{\sum_{m, n} K_G[m, n]}. \quad (2)$$

This Gaussian PSF serves as a reference model to confirm that the CNN framework can learn pre-restoration under a smooth and well-behaved blur kernel.

3.2.2. Fraunhofer Blur

To evaluate AI-based pre-restoration under optics-motivated blur conditions, diffraction-based PSFs are generated using the Fraunhofer diffraction principle[11,12]. Under the Fraunhofer approximation, the intensity distribution in the far field is proportional to the squared magnitude of the Fourier transform of the aperture function $A(x, y)$ [11]. The diffraction PSF is therefore modeled as

$$\text{PSF}(f_x, f_y) = |\mathcal{F}\{A(x, y)\}|^2, \quad (3)$$

where $\mathcal{F}\{\cdot\}$ denotes the 2D Fourier transform and (f_x, f_y) represent spatial frequency coordinates. In this study, the resulting diffraction intensity distribution is discretized to form a PSF kernel $K_D[m, n]$, which is normalized as

$$K_{D,\text{norm}}[m, n] = \frac{K_D[m, n]}{\sum_{m, n} K_D[m, n]}. \quad (4)$$

By selecting different aperture functions $A(x, y)$, various physically interpretable blur kernels can be generated while maintaining strict analytical control[11].

For a circular aperture with radius R , the aperture function is defined as

$$A_{\text{circ}}(x, y) = \begin{cases} 1, & x^2 + y^2 \leq R^2, \\ 0, & \text{otherwise.} \end{cases} \quad (5.1)$$

This model produces an Airy-pattern-like diffraction PSF and provides a representative reference for rotationally symmetric diffraction blur as shown in Fig. 9 as its logarithmic scale using cmap=veridis.

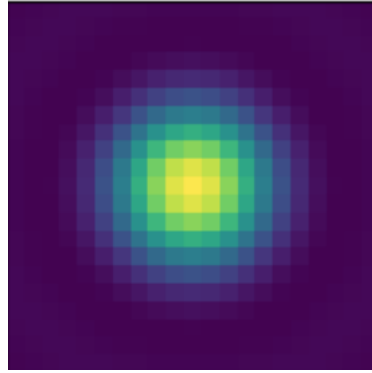


Fig. 9 Circular Fraunhofer PSF

For an equilateral triangle aperture with side length a , centered at the origin and rotated by angle θ , the aperture function is defined by an indicator function

$$A_{\text{tri}}(x, y) = \begin{cases} 1, & (x, y) \in T(a, \theta), \\ 0, & \text{otherwise,} \end{cases} \quad (5.2)$$

where $T(a, \theta)$ denotes the set of points inside the rotated triangular region. This model produces characteristic directional diffraction features and is used to examine whether the CNN can learn pre-restoration for anisotropic, shape-dependent blur. Because triangular apertures are closely related to the geometry of micropism-based retro-reflectors, this model is the most similar to measured AIRR PSFs[21], as shown in Fig. 10 as logarithmic scale using cmap=veridis.

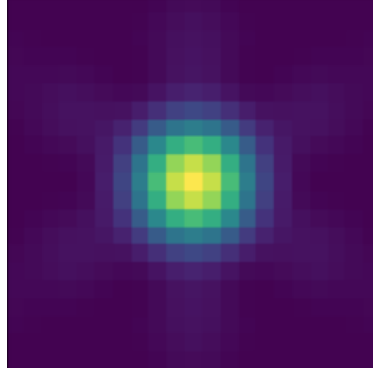


Fig.10 Triangular Fraunhofer PSF

3.2.3. CNN Framework and Architecture

In this study, a convolutional neural network (CNN) is employed as a pre-restoration operator for numerically blurred images, motivated by application to AIRR image formation. Unlike conventional image restoration approaches that attempt to remove blur from a degraded observation, the CNN is designed to generate a pre-corrected image that compensates for the known optical blur in advance. The output of the CNN is therefore intended to be subsequently blurred by the AIRR optical system, or its numerical PSF model, such that the resulting image reproduces the desired target image. This formulation follows the pre-restoration concept proposed for AIRR systems in prior analytical studies [16]. Figure. 11 shows the overall flow of the Model. An input image is first processed by the CNN to generate a pre-corrected image, which is then blurred by a known PSF to simulate AIRR image formation; through training, the CNN learns to produce a pre-corrected output such that the blurred result reproduces the target image.



Fig. 11 CNN General Flow

Let I denote the target image, which is also the original image used as the input to the CNN. The CNN, represented as a nonlinear function $f_{\theta}(\cdot)$ with trainable parameters θ , produces a pre-corrected image $f_{\theta}(I)$. A forward blur operator $H(\cdot)$, modeled as a

convolution with a known PSF, is then applied to this output to simulate the image formation process of the AIRR system. Training of the CNN is formulated as the following optimization problem,

$$\min_{\theta} L(H(f_{\theta}(I)), I), \quad (6)$$

where the CNN parameters are optimized such that the simulated image after blur matches the target image as closely as possible.

In the present work, CNN training is performed in two stages. In the first stage, the loss function L_{MSE} is defined using the mean squared error (MSE) between the blurred CNN output and the target image. MSE is widely used as a basic loss function for image restoration and regression-based learning tasks due to its simplicity and stability [17]. Specifically, the loss is given by

$$L_{\text{MSE}} = \frac{1}{N} \sum_{x,y,c} \left(H(f_{\theta}(I))(x, y, c) - I(x, y, c) \right)^2, \quad (7)$$

where (x, y) denote spatial coordinates, c represents the color channel, and N is the total number of pixels over all channels. By evaluating the error after application of the forward blur operator, the CNN is trained to anticipate and compensate for the optical blur.

In the second training stage, a hybrid loss formulation is employed to further refine the pre-restoration performance. This hybrid loss combines the MSE term with additional structural and perceptual similarity terms, including a structural similarity (SSIM) term and a perceptual similarity term, evaluated after application of the forward blur operator. The purpose of introducing the hybrid loss is to improve structural consistency and training stability during fine-tuning; the specific loss composition is not treated as a contribution of this study.

The hybrid loss L_{hyb} is defined as a weighted sum of three loss components:

$$L_{\text{hyb}} = \alpha L_{\text{MSE}} + \beta L_{\text{SSIM}} + \gamma L_{\text{LPIPS}}, \quad (8)$$

where L_{MSE} is the mean squared error defined in Eq. (4). The SSIM-based loss term is defined as

$$L_{\text{SSIM}} = 1 - \text{SSIM}(H(f_{\theta}(I)), I), \quad (8.1)$$

which penalizes structural dissimilarity between the blurred CNN output and the target image. The perceptual loss term is defined using a deep feature-based distance

$$L_{\text{LPIPS}} = \text{LPIPS}(H(f_{\theta}(I)), I), \quad (8.2)$$

where LPIPS denotes a learned perceptual similarity metric computed from intermediate feature representations of a pretrained convolutional network. The weighting coefficients α, β and γ control the relative contribution of each loss term and are fixed during fine-tuning.

The CNN adopts a compact U-Net-like encoder-decoder architecture designed for image-to-image pre-restoration as shown in Fig. 12. Encoder-decoder-type CNN architectures have been widely applied to image-to-image restoration problems because they enable multiscale feature extraction while preserving spatial correspondence [18]. The network consists of three resolution levels, in which the encoder progressively reduces spatial resolution while increasing the number of feature channels, and the decoder symmetrically restores the spatial resolution. Each encoder stage consists of convolutional processing with downsampling, while the decoder restores spatial resolution through upsampling and convolution to reconstruct spatial details. Skip connections are introduced between corresponding encoder and decoder stages by concatenating feature maps, enabling the preservation of fine spatial information across resolution levels. In addition, a residual connection is applied such that the network predicts a correction term that is added to the input image, allowing the CNN to focus on learning the pre-restoration component required to compensate for the AIRR blur while maintaining overall image structure.

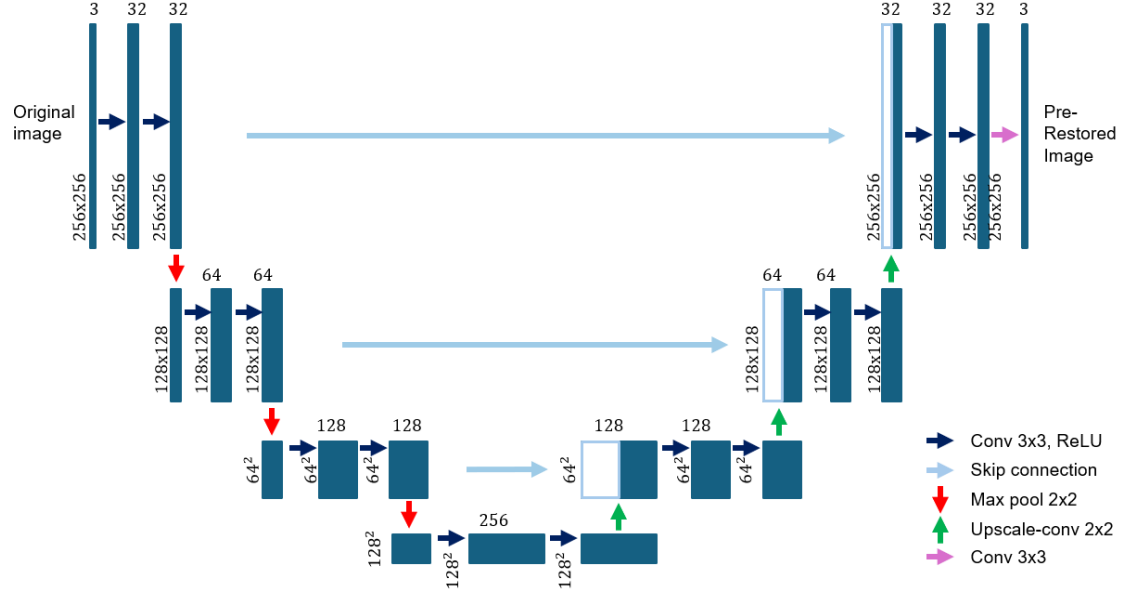


Fig. 12 U- Net Architecture

Through this formulation, the CNN functions as a data-driven pre-restoration module that operates in conjunction with a known PSF model. The framework enables direct evaluation of whether AI-based pre-restoration is feasible under controlled and physically valid blur conditions, as a preliminary step prior to addressing experimentally measured PSFs and their associated noise and artifacts.

3.2.4. Experimental Environment

All numerical experiments in this study were conducted in a controlled software and hardware environment to ensure reproducibility and stable training behavior. The implementation was performed using Python with a dedicated virtual environment (venv) to manage package versions and dependencies [22]. Interactive development, debugging, and visualization were carried out using Jupyter Notebook, which enables iterative inspection of intermediate results and facilitates experimental verification [23].

The convolutional neural network described in the previous section was implemented using the PyTorch deep learning framework. PyTorch provides flexible tensor operations and automatic differentiation, making it suitable for implementing custom image restoration pipelines and training procedures [24]. GPU acceleration was enabled through NVIDIA CUDA to reduce training time and support efficient convolutional operations on high-resolution image data [25].

To improve experimental consistency, fixed random seeds were applied during training for pseudo-random number generators used in network initialization and data handling. This practice reduces run-to-run variation and supports fair comparison across different PSF models, although complete determinism cannot be guaranteed due to nondeterministic GPU operations [26].

The CNN architecture employed in this study follows an encoder–decoder structure inspired by established convolutional network designs for image processing. While the present work does not directly adopt classification-oriented networks, early CNN architectures such as AlexNet demonstrated the effectiveness of deep convolutional structures for learning hierarchical image representations and motivated subsequent developments in image-to-image learning frameworks [27].

Training and inference were executed on a desktop workstation equipped with a consumer-grade GPU. The main hardware specifications of the system used in this study are summarized in Table 1. These specifications are sufficient for conducting feasibility-level experiments involving moderate network depth and image resolution.

Table 1. Hardware Specifications

Hardware	Name
CPU	13thGen Intel® Core™i9-13900 2.00 GHz
GPU	NVIDIA GeForce RTX 4060 Ti
RAM	64 GB 3200 MT/s
SSD	WD Red SN700 2000GB

3.2.5. Experimental Results

This section presents experimental results of CNN-based pre-restoration using synthetic PSF models. Representative results are shown for Gaussian, circular Fraunhofer, triangular, and diamond-shaped PSFs.

For the Gaussian PSF, the training loss decreases smoothly and converges across both training stages, indicating that the CNN can learn a stable pre-restoration mapping under smooth, isotropic blur conditions. Figure 13(a) shows the training loss during the first stage using the MSE-based loss, while Fig. 13(b) shows the loss during the second-stage fine-tuning using the hybrid loss. In both stages, no divergence or oscillatory behavior is observed,

confirming stable optimization behavior.

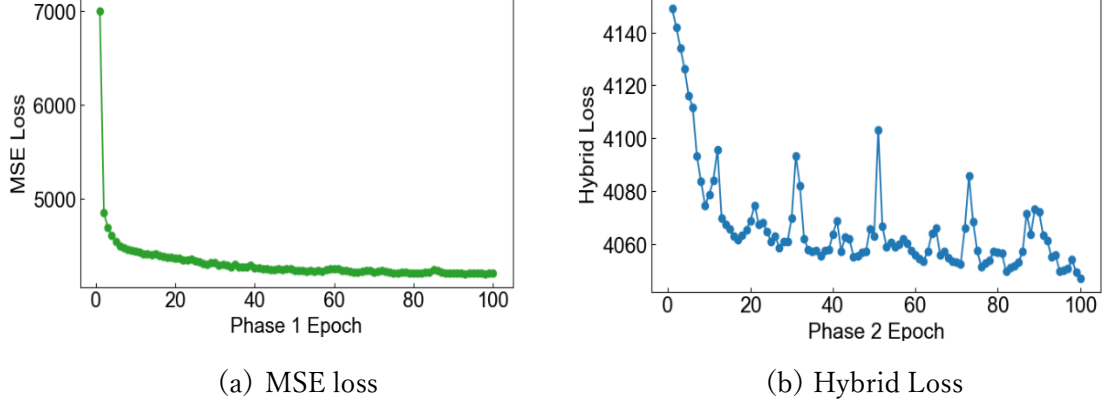


Fig.13 Gaussian PSF training loss

Pre-restoration performance was evaluated across a range of blur strengths by varying the standard deviation σ from 5 to 10 pixels. The corresponding numerical values of PSNR and SSIM for each σ are summarized in Fig. 114 (a) and (b) respectively. For all tested values of σ , the improved blurred images consistently achieve higher PSNR and SSIM than the original blurred images, except σ values 7 and 8, where there was an unexpected drop of SSIM.

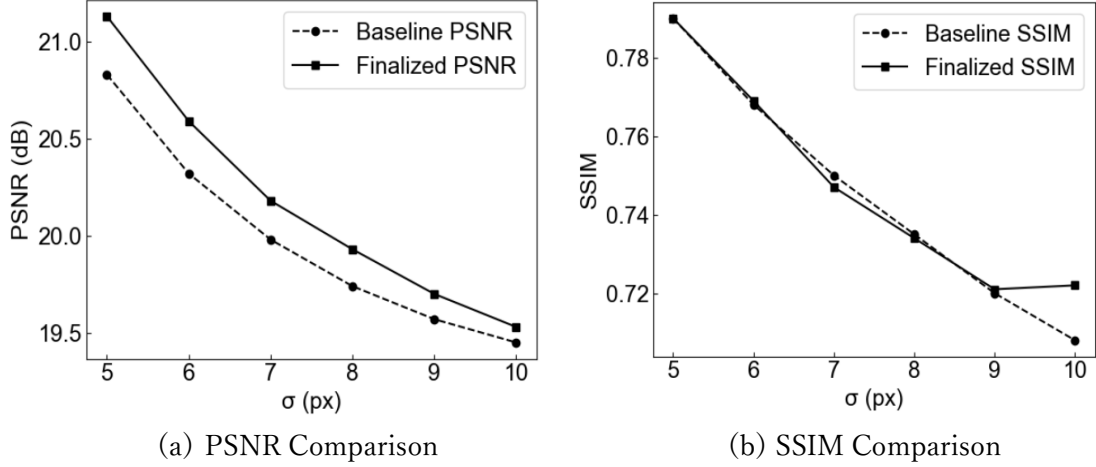


Fig. 14 Gaussian Image Values

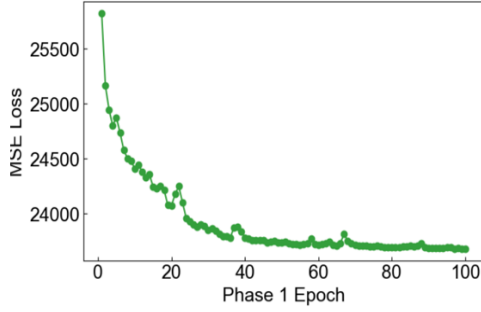
Representative image comparisons for the Gaussian PSF with $\sigma = 7$ are presented in Fig. 15. In the figure, four images are shown: (a) the original image used as the target input I , (b) the blurred image $H(I)$ obtained by applying the Gaussian PSF without pre-restoration, (c) the pre-restored image generated by the CNN $f_{\theta}(I)$, and (d) the improved blurred image

$H(f_\theta(I))$ obtained by applying the same PSF to the pre-restored image.

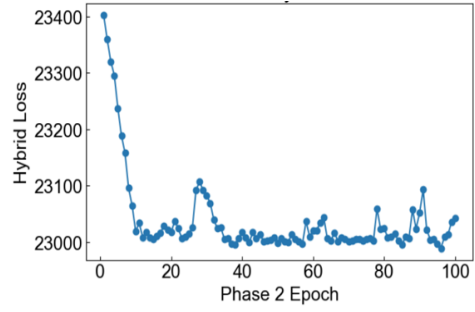


Fig. 15 Gaussian Image Comparisons

Compared with the Gaussian PSF, diffraction-based PSFs introduce structured and oscillatory blur components, resulting in a more challenging inverse problem. Nevertheless, stable convergence is observed in both training stages. Figures 16(a) and 16(b) show the MSE-based and hybrid-loss training curves, respectively, for the circular Fraunhofer PSF, while Figs. 17(a) and 17(b) show the corresponding loss curves for the triangular Fraunhofer PSF. Despite the increased complexity of diffraction-induced blur, the CNN maintains stable learning behavior, and subsequent quantitative metrics indicate consistent improvement over the uncorrected aerial images.

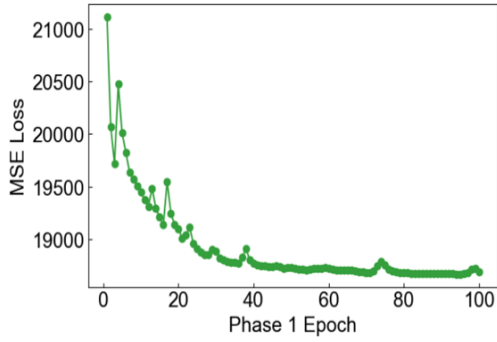


(a) MSE loss

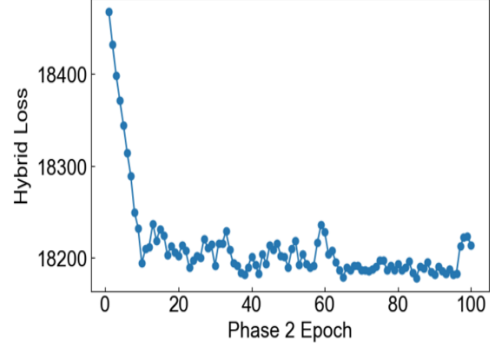


(b) Hybrid Loss

Fig.16 Circular Fraunhofer PSF training loss



(a) MSE loss



(b) Hybrid Loss

Fig. 17 Triangular Fraunhofer PSF training loss

Following Gaussian blur, pre-restoration performance was evaluated using diffraction-inspired PSF models, including circular Fraunhofer and triangular Fraunhofer PSFs. Representative image comparison results for the circular Fraunhofer PSF, and triangular PSF are shown in Fig. 18, and 19 respectively. For each PSF, the same four-image layout is used, (a) original image I , (b) blurred image $H(I)$, (c) pre-restored image $f_{\theta}(I)$, and (d) improved blurred image $H(f_{\theta}(I))$. The corresponding PSNR and SSIM values for each PSF type are summarized in Tables 2, and 3.

Table 2. Circular Fraunhofer Results

Image	PSNR(dB)	SSIM
Original Blurred Image	22.10	0.8654
Improved blurred Image	23.90	0.8766



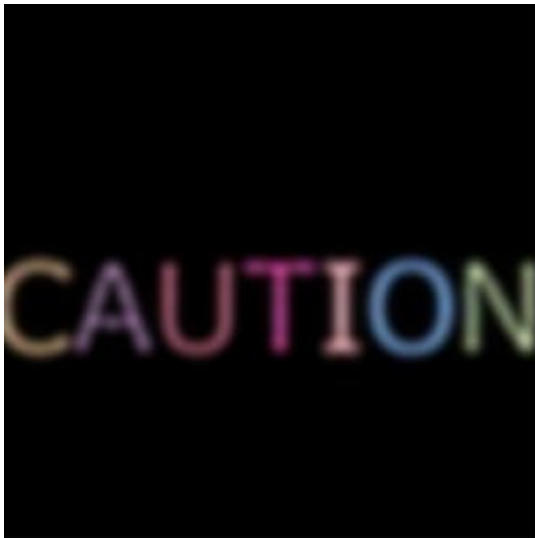
(a) Original Image



(b) Original Blurred Image



(c) Pre-Restored Image



(d) Improved Blurred Image

Fig. 18 Circular Fraunhofer Image Comparisons

Table 3. Triangular Fraunhofer Results

Image	PSNR(dB)	SSIM
Original Blurred Image	24.05	0.8771
Improved blurred Image	25.27	0.8895

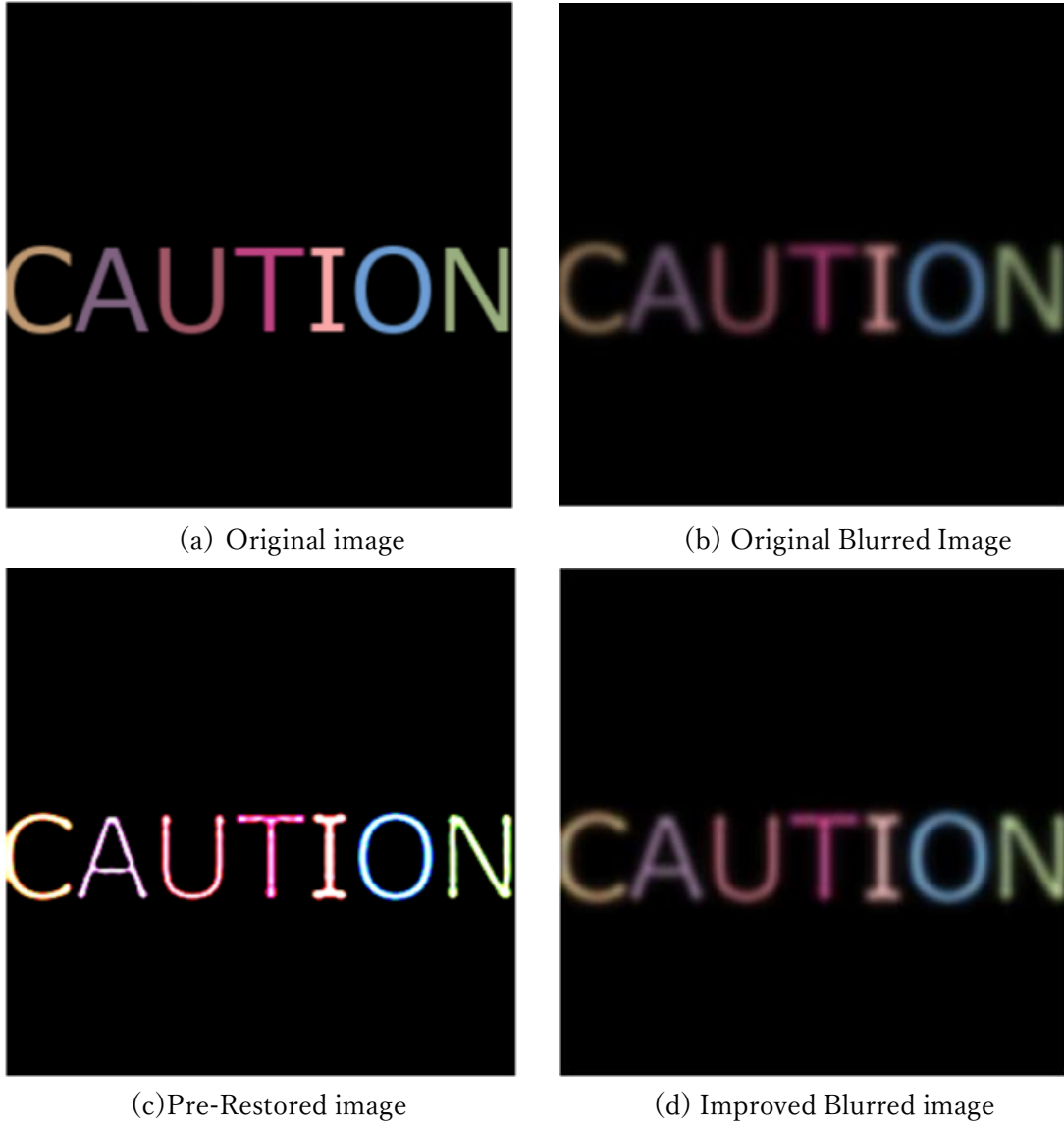


Fig. 19 Triangular Fraunhofer Image Comparisons

Across all tested PSF models, the blurred images formed without pre-restoration differ from the original image due to the applied blur. After applying pre-restoration, the resulting blurred images show small but consistent differences compared with the original blurred images. Quantitative evaluation using PSNR and SSIM, computed between the blurred images and the original image, indicates higher values for the improved blurred images than for the original blurred images for all PSF types. These results indicate that the proposed CNN-based framework can produce measurable improvements under synthetic blur conditions, supporting its feasibility as a pre-restoration approach.

3.3 Summary

This chapter investigated the feasibility of AI-based pre-restoration for AIRR systems using synthetic PSF models. A CNN-based framework was formulated to generate pre-corrected images that are evaluated after numerical blur, representing simulated image formation. Experimental results using multiple synthetic PSF types showed small but consistent improvements in the resulting blurred images, as confirmed by quantitative evaluation. These results demonstrate that AI-based pre-restoration is feasible under controlled, simulated blur conditions, within the limited scope of this study.

4 PSF Measurement in AIRR Systems and Pedestal Artifacts in AIRR

4.1 Introduction

Chapter 2 introduced the point spread function (PSF) as a fundamental descriptor of image degradation in AIRR systems. The PSF represents the spatial response of the optical system to a point input and provides a physically meaningful basis for numerical image simulation, resolution analysis, and computational image restoration [11]. Consequently, accurate PSF characterization is essential for reliable analysis of AIRR image quality. In the previous chapter, PSFs were assumed to be analytically defined and physically valid, enabling controlled evaluation of image degradation and AI-based pre-restoration. In contrast, PSFs in practical AIRR systems must be obtained experimentally, and such measured PSFs often include additional non-physical components introduced by the measurement process [28].

Unlike conventional imaging systems, AIRR systems do not form images on a physical image plane, which makes direct PSF measurement difficult. Because aerial images are formed in free space, PSF acquisition relies on indirect measurement methods using cameras and auxiliary optical setups [28]. As a result, experimentally acquired PSFs often include components originating from the measurement system rather than from the intrinsic optical response of the AIRR system.

Measured AIRR PSFs frequently exhibit background pedestal components, which appear as non-zero baseline intensity distributed over the image. These components can arise from sensor offset, dark current, background illumination, and stray light in the measurement environment [29]. Although such pedestal components may appear insignificant in linear-scale images, they can strongly distort PSF shape when used for numerical convolution or deconvolution, leading to non-physical blur modeling and unreliable image processing results

[30].

This chapter focuses on the measurement of PSFs in AIRR systems and the analysis of pedestal artifacts contained in experimentally acquired PSFs. The origin and characteristics of pedestal components are examined, and a practical preprocessing method based on pedestal subtraction is introduced to restore the physical validity of the PSF while preserving its intrinsic structure [16]. This preprocessing framework enables reliable use of measured PSFs for image simulation and restoration in subsequent chapters.

4.2 Experiment

4.2.1. PSF Measurement Method for Aerial Images

In AIRR systems, the point spread function (PSF) cannot be measured directly on a physical image plane because the image is formed in free space [7,8]. To address this limitation, Watanabe et al. proposed an indirect PSF measurement method based on capturing an aerial point image using a camera system [31]. This study follows the same measurement approach to ensure consistency with established PSF measurement methodology for plane-symmetric real-image-forming aerial display optics.

The experimental setup used for PSF acquisition is shown in Fig. 20. A halogen light source is employed as the illumination source, and its output is first coupled into an integrating sphere to generate spatially uniform illumination. An aperture stop is placed at the output of the integrating sphere to define a small circular light source, which serves as an approximate point impulse. By changing the aperture diameter, point sources with diameters ranging from 1 mm to 6 mm can be generated.

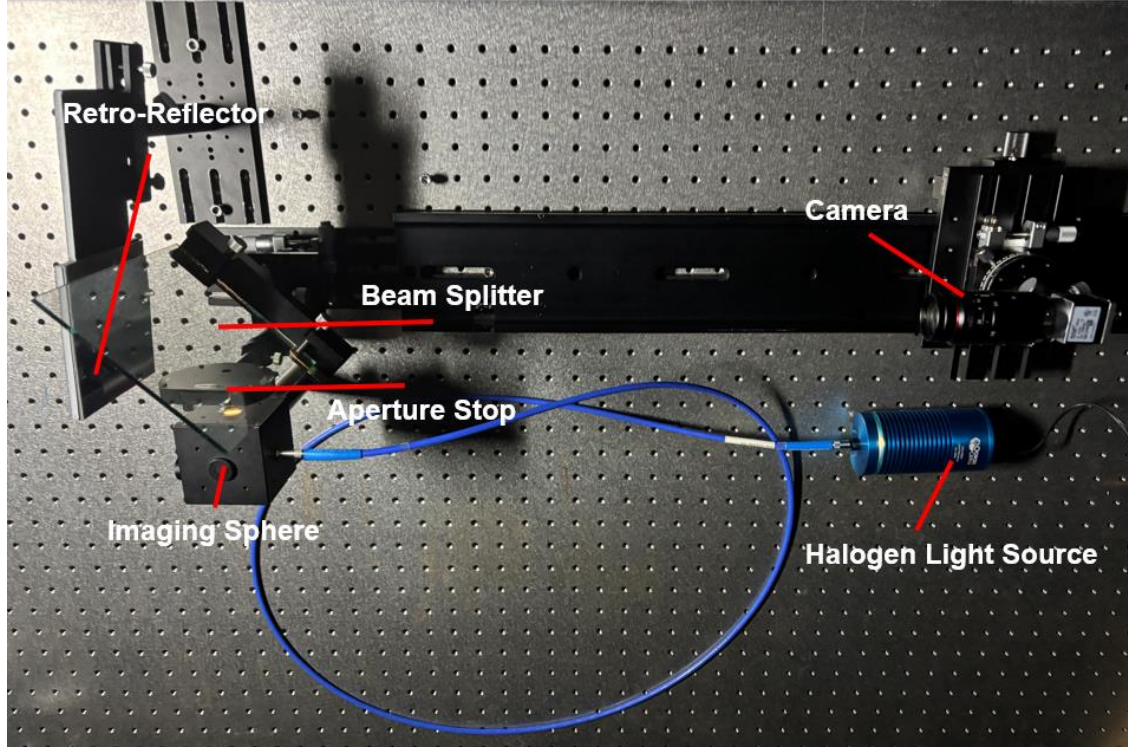


Fig. 20 PSF acquisition experimental setup

The generated point light is introduced into the AIRR optical system, which consists of a half mirror (beam splitter) and a microprism-type retro-reflector [7]. Light emitted from the point source is reflected by the half mirror and directed toward the retro-reflector. The retro-reflected light returns to the half mirror, where it is transmitted toward the observation side, forming an aerial image in free space at a position plane-symmetric to the light source [7].

The aerial point image is captured using an optical camera positioned at the aerial image plane. In this study, a Basler industrial camera is used to record the spatial intensity distribution of the aerial image. The camera is mounted on a translation stage to accurately align the sensor plane with the aerial image plane [28]. All measurements are conducted under controlled lighting conditions to reduce the influence of ambient illumination.

The captured image of the aerial point source represents the system response to a localized input and is treated as the experimentally acquired PSF [11]. For subsequent numerical analysis, a region of interest (ROI) centered on the intensity peak is extracted from the captured image. Although this measurement method enables practical acquisition of AIRR PSFs, the recorded data inevitably include contributions from the measurement system itself,

such as sensor offset and background illumination [29]. These effects appear as additive background components in the measured PSF and are examined in detail in the following section.

4.2.2. Background Pedestal Components

Experimentally acquired PSFs in AIRR systems often contain background intensity components that are not part of the intrinsic optical impulse response [31]. These components originate from the PSF measurement process itself rather than from diffraction or geometrical effects within the AIRR optical system. In this study, such background components are referred to as pedestal components.

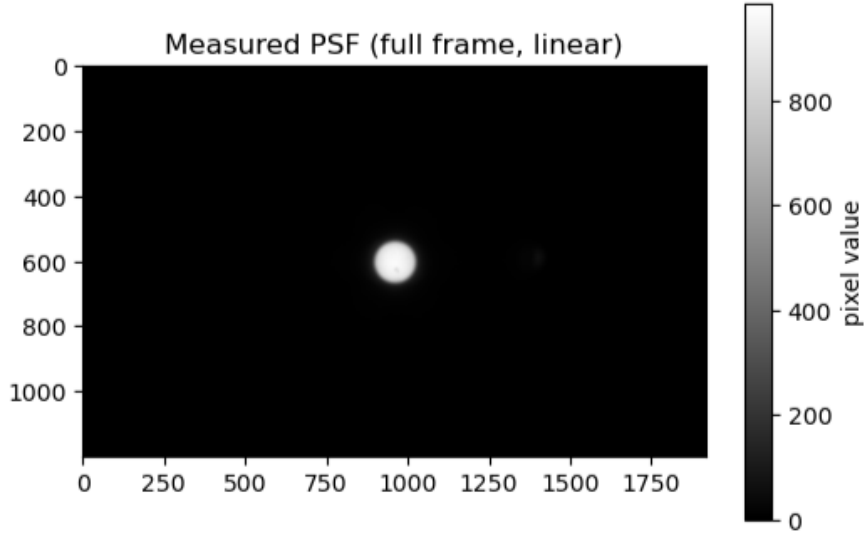
The measured PSF image $I(x, y)$ is modeled in equation (9),

$$I(x, y) = P(x, y) + b + n(x, y) \quad (9)$$

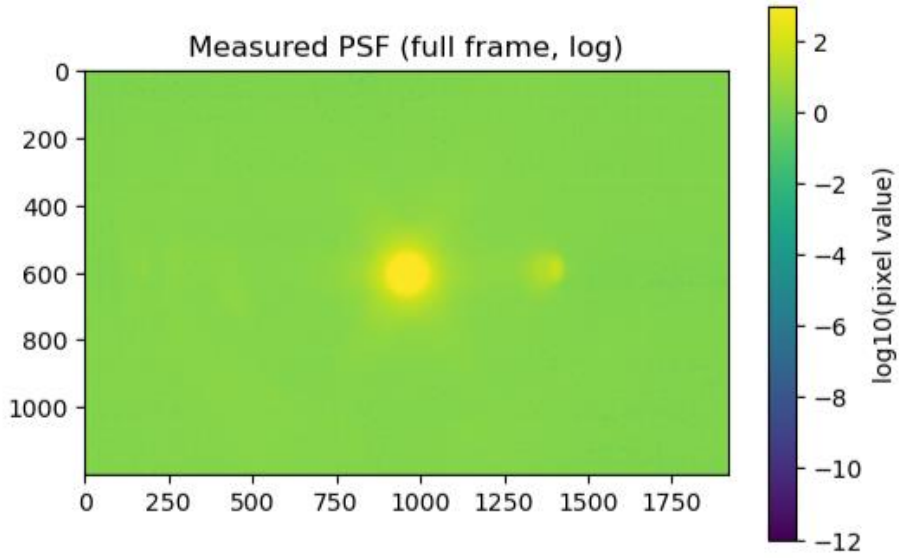
where $P(x, y)$ denotes the intrinsic PSF intensity of the optical system, b represents a pedestal component, and $n(x, y)$ corresponds to random noise [29]. The pedestal component b is assumed to be spatially uniform within the region of interest (ROI) and originates from sensor bias, dark current, and background illumination[29]. Random noise $n(x, y)$ is treated as a zero-mean stochastic fluctuation and is not explicitly removed in this preprocessing.

The objective of the preprocessing described below is to estimate and remove the additive pedestal component b from the measured PSF image in order to obtain a physically consistent PSF suitable for numerical blur simulation and image restoration.

Fig. 21(a) shows the raw measured AIRR PSF represented as a two-dimensional luminance distribution on a linear intensity scale. On this scale, the PSF appears to be dominated by a localized main lobe corresponding to the system's response to a point input. However, when the same data are visualized on a logarithmic scale, as shown in Fig. 21(b), a non-zero background level distributed over the entire image becomes clearly visible. This background pedestal persists far outside the main lobe region and does not decay toward zero intensity.



(a)Raw Data



(b)Logarithmic Scale

Fig. 21 Raw measured PSF and logarithmic visualization

The presence of pedestal components can be attributed to several measurement-system-induced effects, including sensor bias and offset, dark current, background illumination, and stray light within the optical setup [29]. These effects introduce an additive intensity offset that is independent of the AIRR system's optical response. Although such offsets may be negligible in conventional imaging contexts, they become problematic when PSFs are used as convolution kernels for numerical image blur simulation or

restoration[15,30].

To facilitate analysis, the measured PSF is cropped to a fixed 401x401 region of interest (ROI) centered at the dominant intensity peak, as described in Section 3.2. The extracted ROI and its corresponding log-scale visualization are shown in Fig. 22(a) and Fig. 22(b) respectively. Notably, the pedestal component remains present within the ROI and appears as a uniform baseline surrounding the main lobe. This demonstrates that ROI cropping alone is insufficient to remove measurement-induced background components.

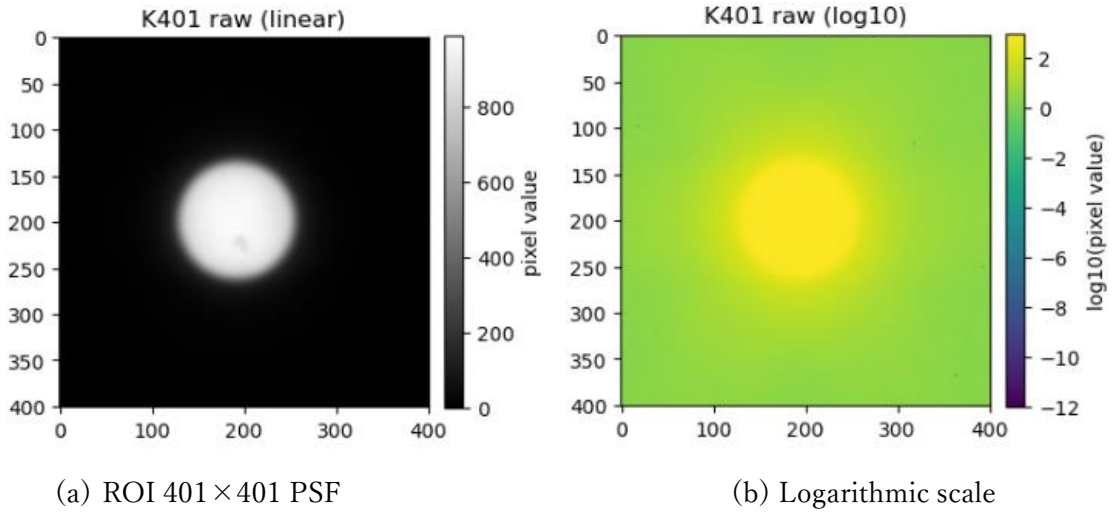


Fig. 22 ROI cropping

To estimate the additive pedestal component b , a background estimation region is defined within the cropped PSF kernel. A border-ring region Ω_{ring} of width 10 pixels surrounding the outer boundary of the 401 × 401 ROI is selected. This region is sufficiently distant from the PSF core such that contributions from the intrinsic PSF are expected to be negligible [11]. Log-scale visualization of the raw PSF confirms that the intensity distribution in this outer region converges to a spatially uniform baseline, consistent with the presence of an additive pedestal component.

The pedestal value b is estimated as the median intensity within the border-ring region:

$$b = \text{median}(K_{\text{raw}}(x, y) \mid (x, y) \in \Omega_{\text{ring}}). \quad (10)$$

The median is employed as a robust statistic to reduce sensitivity to outliers caused by residual noise or localized artifacts[14,29]. This approach follows standard background

estimation practices in image processing, where additive offset components are estimated from image regions assumed to contain negligible signal contribution[14].

Table 4 summarizes the properties of the measured PSF and the extracted ROI. Although the maximum intensity within the ROI is significantly larger than the background level, the pedestal value remains non-zero and comparable across the cropped region. As a result, the measured PSF violates key physical assumptions required for PSF-based image formation models. A physically valid PSF should decay toward zero away from the optical axis and conserve energy when used as a convolution kernel [11,12,13].

Table 4. Measured PSF properties & pedestal statistics

Data	Value
Full PSF size	1200×1920
COM location	(604, 968)
ROI size	401×401
ROI max	987
Pedestal (median)	2.7

If PSFs with pedestal artifacts are directly used for numerical convolution, the pedestal component introduces artificial global intensity offsets and distorts the effective blur characteristics[15,16]. These effects compromise the physical interpretability of blur simulation results and can adversely affect subsequent image processing or restoration algorithms.

Therefore, appropriate treatment of pedestal components is essential before measured AIRR PSFs can be used for physically meaningful image simulation and analysis. The following section introduces a practical preprocessing method to estimate and remove pedestal components while preserving the intrinsic spatial structure of the measured PSF.

4.2.3. Pedestal Subtraction Method

Based on the pedestal value b estimated in the previous section, pedestal subtraction is applied to the measured PSF kernel following standard background correction practices in image processing [14,29]. The pedestal subtracted PSF kernel $K_{ps}(x, y)$ is defined as

$$K_{ps}(x, y) = K_{raw}(x, y) - b. \quad (11)$$

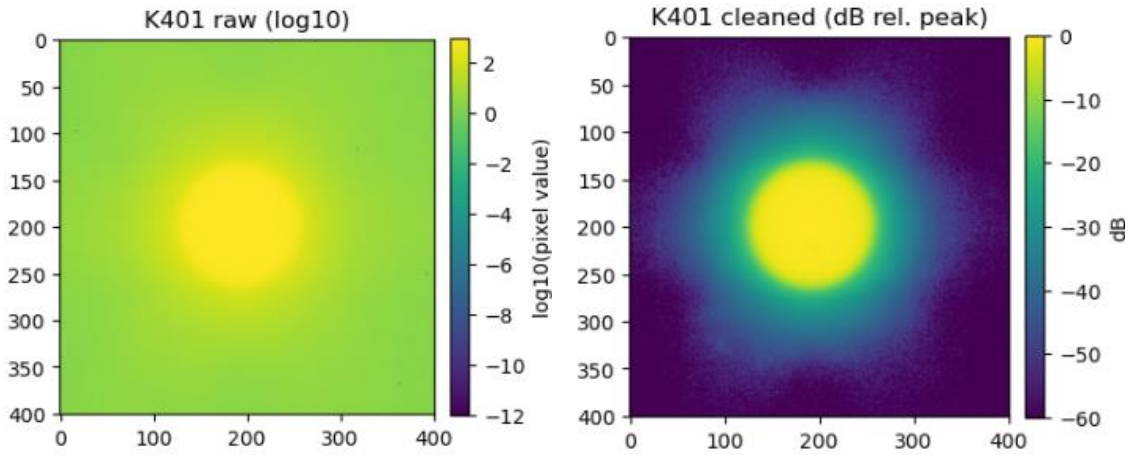
Negative values resulting from the subtraction are clipped to zero, as the PSF represents a non-negative intensity distribution[11].

The pedestal subtracted kernel is then normalized to obtain a physically valid PSF suitable for numerical image processing. The normalized PSF kernel $K_{\text{norm}}(x, y)$ is defined as,

$$K_{\text{norm}}(x, y) = \frac{K_{\text{ps}}(x, y)}{\sum_{x, y} K_{\text{ps}}(x, y)}. \quad (12)$$

This normalization enforces energy conservation and ensures that the processed PSF kernel can be directly used as a discrete convolution kernel for numerical blur simulation and image restoration. The normalized PSF kernel is shifted such that the pixel with maximum intensity coincides with the geometric center of the ROI. This alignment improves numerical stability and consistency in subsequent convolution operations[14].

Figure 23 shows logarithmic-scale visualizations of the PSF kernel before and after pedestal subtraction and normalization. In the raw PSF K_{raw} , a non-zero baseline persists over the entire ROI, which is not apparent in linear-scale representations. After applying pedestal subtraction and normalization, the background level is effectively suppressed, and the PSF intensity decays toward zero away from the core. This confirms that the proposed preprocessing removes non-physical background components while preserving the intrinsic spatial structure of the PSF.



(a) Raw PSF K_{raw}

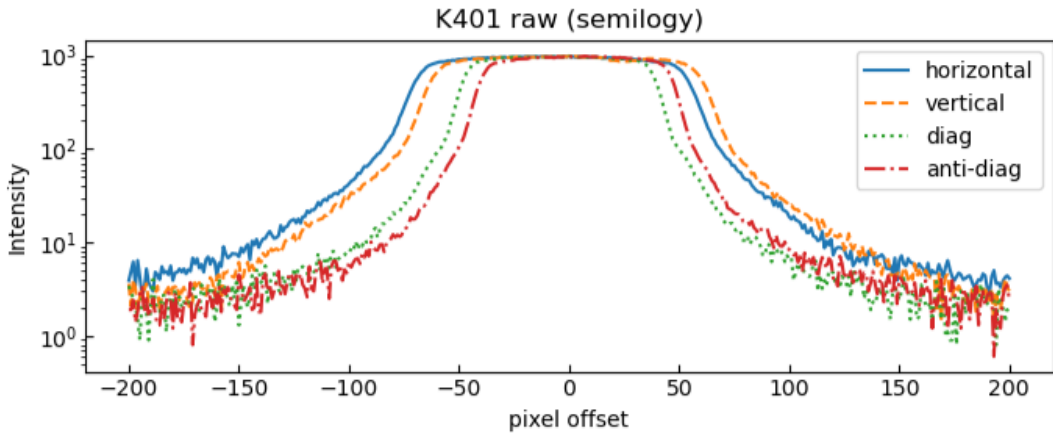
(b) Pedestal Subtracted PSF K_{ps}

(c) Fig 23 PSF comparison after Pedestal Subtraction

4.2.4. Experimental Results

To evaluate the robustness of the pedestal subtraction and normalization procedure, the preprocessing pipeline defined in Sections 4.2.2 and 4.2.3 was applied to multiple experimentally measured PSFs acquired under different aperture conditions. The PSF datasets were provided by Watanabe et al. and correspond to circular aperture diameters ranging from 1 mm to 6 mm [31]. All PSF datasets were processed using identical preprocessing parameters and procedures, including ROI size, pedestal estimation strategy, subtraction, and normalization. No aperture-dependent tuning or parameter adjustment was performed, ensuring a consistent evaluation across all conditions.

Representative logarithmic-scale visualizations of PSF cross-sections before and after preprocessing for a representative aperture diameter of 6 mm are shown in Fig. 24. In the raw measured PSFs shown in Fig. 24(a), a non-zero baseline persists in the low-intensity tail region, indicating the presence of an additive background component that does not decay toward zero. After pedestal subtraction and normalization, this background component is effectively suppressed, and the PSF intensity consistently decays toward zero away from the core for all aperture conditions, with results for other apertures provided in the Appendix as shown in Fig. 24(b).



(a)

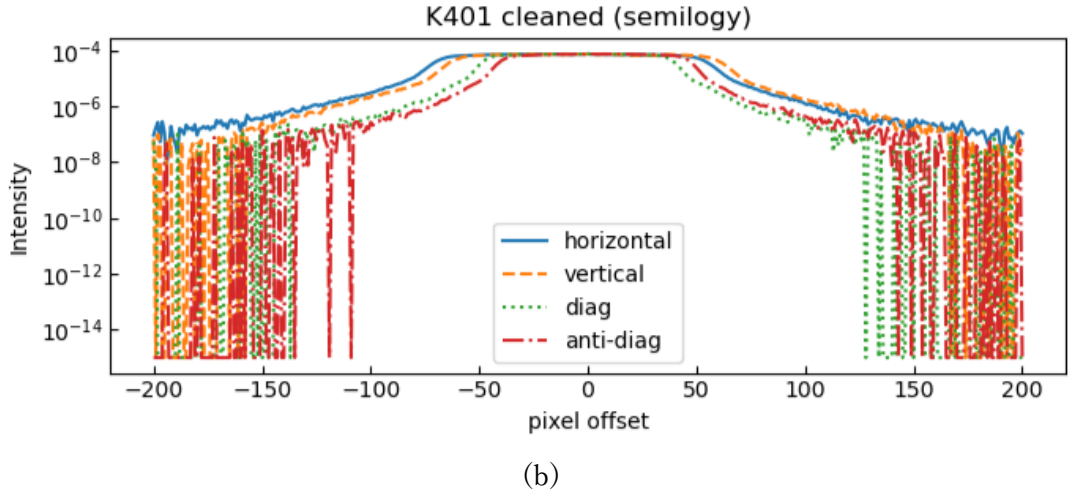
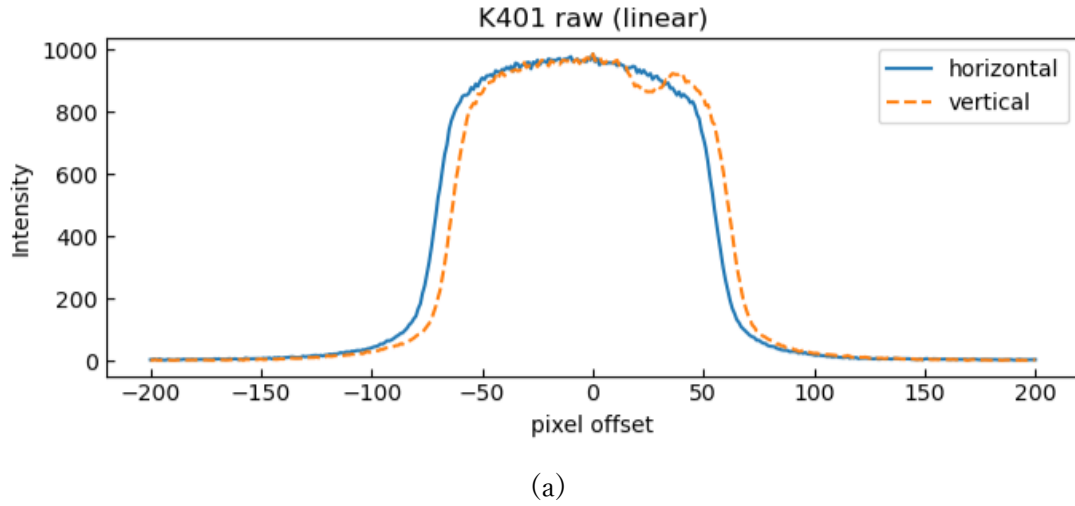


Fig. 24 Logarithmic-scale PSF cross-sections before(a) and after(b) processing

Linear-scale comparisons confirm that the preprocessing does not alter the shape or relative intensity distribution of the PSF core for the representative 6 mm aperture case as shown in Fig. 25(a) and (b). The peak structure and core width remain unchanged, indicating that the observed differences between the raw and processed PSFs are attributed to the removal of the additive pedestal component rather than to distortion or artificial sharpening of the intrinsic optical response.



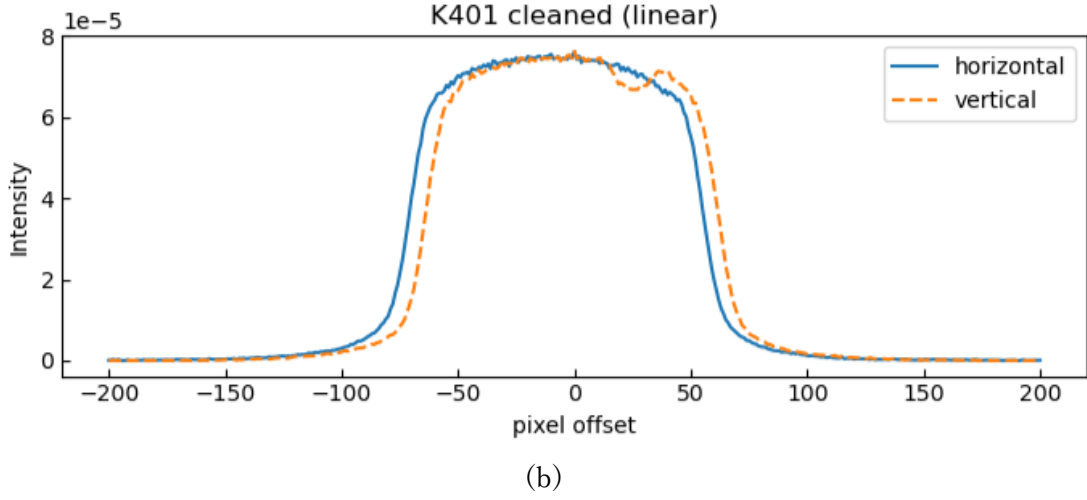


Fig. 25 Linear-scale PSF cross-sections before(a) and after(b) processing

These results demonstrate that the proposed preprocessing pipeline can be consistently applied to experimentally measured PSFs obtained under different aperture conditions, yielding physically valid PSF kernels suitable for numerical blur simulation and subsequent image processing [11].

4.3 Summary

In this chapter, the characteristics of experimentally measured point spread functions (PSFs) in an Aerial Imaging by Retro-Reflection (AIRR) system were examined, with emphasis on measurement-system-induced pedestal components. Because PSF acquisition for aerial images relies on indirect measurement, experimentally measured PSFs were shown to include additive background offsets originating from the imaging sensor and measurement environment rather than from the intrinsic optical response of the AIRR system. Logarithmic-scale analysis revealed that raw measured PSFs exhibit a non-zero baseline in low-intensity regions, indicating the presence of pedestal components that violate the physical requirements of a finite-energy PSF. As a result, raw measured PSFs cannot be directly used for physically meaningful numerical blur simulation or image processing.

To address this issue, a preprocessing pipeline based on pedestal subtraction and normalization was introduced. The pedestal component was estimated from a background region at the periphery of the cropped PSF and removed, followed by normalization to enforce energy conservation. Application of this preprocessing to multiple measured PSFs acquired under different aperture conditions demonstrated consistent suppression of non-physical background components while preserving the intrinsic PSF core structure.

These results confirm that pedestal subtraction is an essential preprocessing step when using experimentally measured PSFs for AIRR image analysis and simulation. The resulting pedestal-subtracted PSFs provide a physically valid representation of system blur and form a reliable foundation for subsequent numerical image degradation and restoration studies.

5 Conclusion

5.1 Thesis Summary

In accordance with the research objective, this thesis conducted a fundamental investigation into the relationship between point spread function (PSF) characterization and AI-based image enhancement in aerial imaging optical systems, with particular focus on AIRR systems. The feasibility study using synthetic and analytically defined PSFs demonstrated that AI-based pre-restoration can achieve measurable improvements when the blur process is physically valid and well controlled. In parallel, analysis of experimentally measured AIRR PSFs revealed that raw measurement data often include non-physical pedestal components originating from the measurement chain, which must be addressed before such PSFs can be reliably used for computational image analysis. Taken together, these results clarify that while AI-based pre-restoration is feasible under simplified conditions, its meaningful application to real AIRR systems fundamentally depends on physically consistent PSF characterization and preprocessing. This establishes a necessary foundation for future PSF-aware AI-based image enhancement that integrates both analytical restoration theory and data-driven approaches.

5.2 Future work

Future work will focus on extending the proposed framework in several directions. One important step is the incorporation of experimentally measured PSFs, after appropriate preprocessing, into the AI training and evaluation pipeline, enabling investigation of pre-restoration under more realistic conditions. This includes assessing the influence of residual measurement artifacts and variability in measured PSFs on AI performance.

Moreover, improvements to the AI architecture and training strategy should be explored. Possible directions include investigating alternative network structures, loss functions, and training schemes that are better matched to the characteristics of optical blur in aerial imaging systems, as well as improving robustness to PSF variation.

Finally, the approach presented in this thesis can be extended beyond AIRR to other forms of aerial and free-space image display technologies. Applying similar PSF-aware and

AI-assisted image enhancement frameworks to different optical configurations may contribute to broader improvements in image quality across a range of aerial imaging systems.

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Appendices

The following Figures 26 to 36 shows the following manuscripts used in this thesis. Figures 30 to 36 represent the semiology graphs for aperture sizes 1mm to 5mm.



Fig. 26 IDW'25 Manuscript 1.

where A denotes the aperture transmission function, F is the two-dimensional Fourier transform, and PSF is the point spread function.

We constructed binary apertures corresponding to different geometries—circular ($x^2+y^2 \leq R^2$), and triangular ($(x,y) \in T(a,\theta)$). Where, R denotes the radius of a circular pupil, a is the side length of an equilateral triangular pupil rotated by angle θ . Each aperture was super sampled for anti-aliased edges, transformed via FFT, and squared in magnitude to obtain the diffraction PSF. The resulting kernels exhibit anisotropy and sidelobe structures (Airy rings for the circle, sixfold spikes for the triangle), which better approximate the real PSFs observed in AIRR optical systems.

3.2 Dataset and Preprocessing

Our dataset consisted of high-resolution sharp images resized to 256×256 px and normalized to [0,1]. Noisy or low-contrast images were excluded. For each image, we generated blurred counterparts within the data loader: after loading and resizing with OpenCV (BGR→RGB conversion), we applied per-channel Gaussian blur with the same σ across R, G, and B, yielding paired (sharp, blurred) tensors. Prior to training, all images were converted to PyTorch tensors with pixel-wise scaling, and we randomized their order each epoch to encourage generalization across scene content.

3.3 Hybrid Loss Function and Two-Phase Training

We trained the model in two distinct phases using a hybrid loss that balances pixel-level fidelity, structural similarity, perceptual realism, and color consistency. In Phase 1, we optimized a pure summed-MSE loss between the re-blurred network output and the sharp target (equivalent to maximizing PSNR), running up to 100 epochs but employing early stopping with a patience of 10 epochs once the training MSE plateaued.

In Phase 2, we fine-tuned the network for perceptual quality using a composite loss. The loss function is defined as

$$L = \alpha MSE + \beta(1 - SSIM) + \delta LPIPS + \gamma L_{color}, \quad (2)$$

where α , β , δ , and γ were ramped linearly over the first 10 epochs of Phase 2 to their final values ($\alpha=0.95$, $\beta=0.03$, $\delta=0.01$, $\gamma=0.01$), ensuring that the sum of weights remained 1.0. The SSIM term was implemented as $(1-SSIM)$ to use as a loss function scaled by the number of pixels. LPIPS was computed with an AlexNet backbone and summed per batch, and the color penalty enforced channel-wise mean consistency via MSE on mean RGB values.

3.4 Implementation Details

All optimization used Adam with an initial learning rate of 1×10^{-3} , a batch size of 32. During Phase 2, we monitored validation composite loss with a patience of 100 epochs, checkpointing the model whenever the combined

loss improved. Training and validation PSNR and SSIM were recorded each epoch, confirming that Phase 1 quickly reached a PSNR plateau and Phase 2 modestly improved perceptual metrics without sacrificing fidelity. In all experiments, spatially differentiable re-blur was performed via grouped 3-channel convolution with the same Gaussian kernels used for data generation.

3.5 Training

Training was conducted in two sequential phases using the synthetic blur dataset starting off with gaussian blur. In Phase 1, the network was trained for 100 epochs to minimize only the mean-squared error (MSE) between the pre-corrected output and the sharp ground truth. The MSE loss dropped precipitously in the first few epochs and then plateaued as shown in Fig. 2(a).

Phase 2 extended training for another 100 epochs under the full hybrid objective, which combined MSE, SSIM, LPIPS, and color-consistency terms. The hybrid loss trajectory decreases steadily after the perceptual and color losses are ramped to their final weights as shown in Fig. 2 (b). We then extended the training using the Fraunhofer synthetic blur. The training data of the same model used by the Gaussian blur is as shown in Fig. 3. The training shows that the model also does adequately towards the Fraunhofer Diffraction blur.

The full 200-epoch training run required approximately 30 hours per training session on a single RTX 4060 Ti GPU.

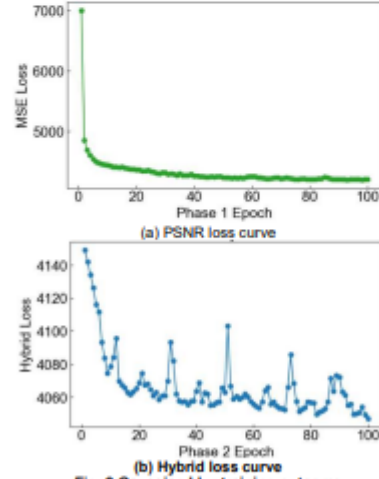


Fig. 2 Gaussian blur training outcome

Fig. 27 IDW'25 Manuscript 2.

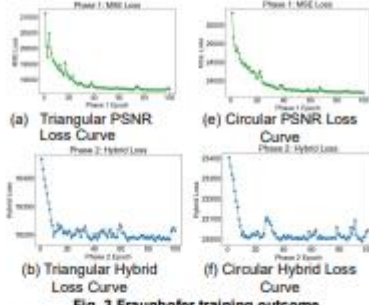


Fig. 3 Fraunhofer training outcome

4 Results

4.1 Quantitative Metrics

This section summarizes PSNR and SSIM for Gaussian blur levels $\sigma = 5\text{--}10$ px, comparing the uncorrected (baseline) images against the U-Net pre-restoration (finalized) outputs. For $\sigma = 5$, PSNR rises from 20.83 dB at baseline to 21.13 dB after pre-restoration. As σ increases, the absolute PSNR gain narrows slightly, yet the network consistently delivers higher fidelity across the entire blur range as shown in Fig. 4. SSIM exhibits a small but systematic improvement for intermediate sigmas before converging again at larger blurs as shown in Fig. 5. These results demonstrate that a single model trained $\sigma = 5$ px generalizes to stronger blurs up to $\sigma = 10$ px, yielding net gains in both pixel-level accuracy and structural similarity without any per-sigma fine-tuning.

The comparison for the Fraunhofer Diffraction is as shown in Table 1. The triangular aperture has a baseline PSNR of 24.01dB that increases to 25.27dB through the model. The SSIM for the triangular blur also increases from the baseline 0.8771 to the finalized 0.8895.

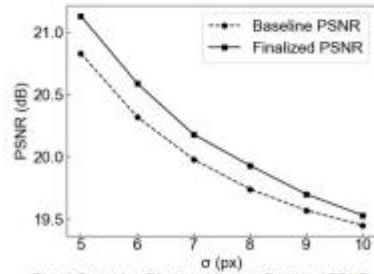


Fig. 4 Gaussian Blur baseline vs finalized PSNR comparison

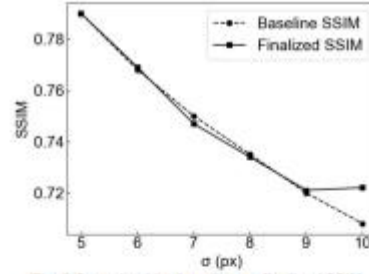


Fig. 5 Gaussian blur baseline vs finalized SSIM comparison

Table 1 Fraunhofer Results

	Triangular	Circular
Baseline PSNR (dB)	24.01	22.10
Finalized PSNR (dB)	25.27	23.90
Baseline SSIM	0.8771	0.8654
Finalized SSIM	0.8895	0.8766

4.2 Qualitative Metrics

In this section we compare 4 images using a multicolor caution sign generated using python. The Gaussian blur kernel used is blur $\sigma = 7$ px. The original sharp image (a), the baseline blur (b), the pre-restored image (c) and the finalized re-blur (d) is shown in Fig. 6. We also compare another 4 images from the Fraunhofer triangular blur training session as shown in Fig. 7 and circular blur in Fig. 8.

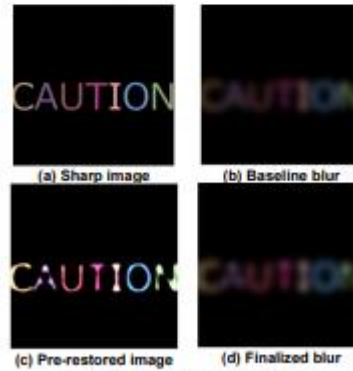


Fig. 6 Gaussian blur comparison



Fig. 7 Triangular blur comparison



Fig. 8 Triangular blur comparison

5 Discussion

The quantitative results of the model trained on Gaussian PSFs ($\sigma = 5\text{--}10$ px) show consistent, though modest, improvements in both PSNR and SSIM across blur levels. A PSNR gain of ~ 0.30 dB for $\sigma = 5$ px, while small, is meaningful in pre-restoration for mild blur, where the aim is to recover sharpness without introducing artifacts such as ringing, a common issue in classical deconvolution. Notably, this advantage persists at larger σ values ($\sigma = 10$ px), suggesting the network's inverse mapping generalizes beyond the mild blurs seen during training. SSIM improvements are smaller but steady, reflecting the trade-off between pixel fidelity and structural similarity. This arises from the loss design, dominated by MSE ($\alpha = 0.95$) with only a weak SSIM term ($\beta = 0.03$) and

the re-blur validation scheme, where the output remains inherently blurry and structurally close to the baseline. Even so, the observed SSIM gains confirm the model captures perceptual improvements within these constraints.

A key limitation is the simplified blur model. The true AIRR PSF is anisotropic and spectrally variant across color channels, but our current training used isotropic Gaussian blur applied identically to R, G, and B. This proved the viability of pre-restoration but does not capture the system's real complexity. To bridge this gap, we introduced synthetic physically derived PSFs from Fraunhofer diffraction of circular and triangular apertures. These exhibit anisotropy and sidelobes (e.g., Airy rings, diffraction spikes), making them more representative of AIRR blur and providing a systematic test of network robustness.

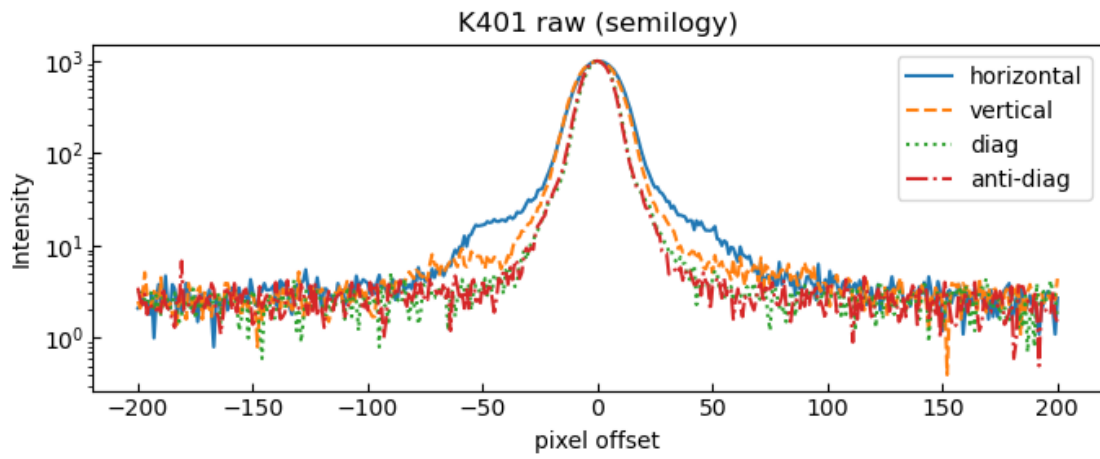
Future work will move beyond synthetic kernels to training with measured, channel-specific PSFs from the AIRR system. This would allow the model to learn the true color-dependent blur and adapt to optical variations. Broader directions include training on diverse real PSFs to improve generalization or embedding PSF learning into a blind deconvolution framework. Ultimately, validating the pre-restoration model in a real-time AIRR setup will provide the decisive test of this data-driven approach.

6 Conclusions

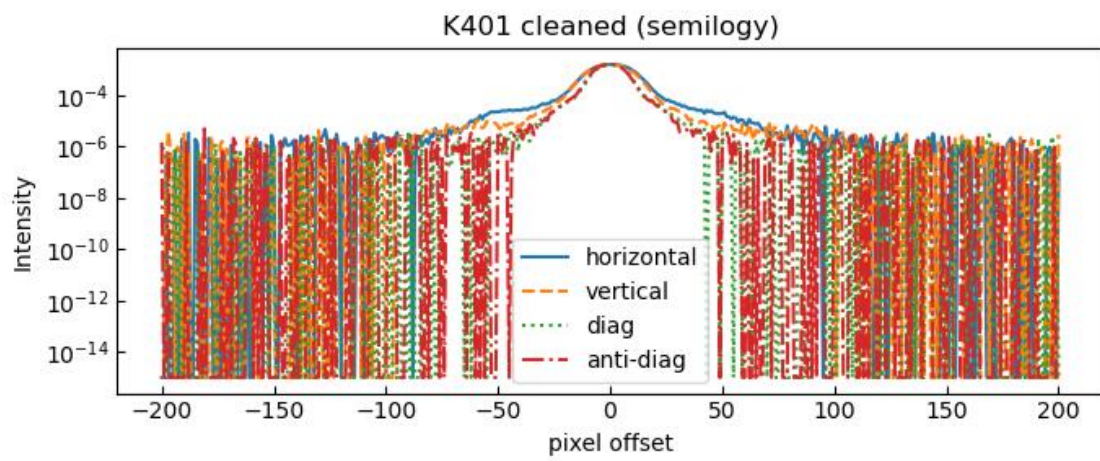
We have developed a U-Net for pre-restoring the mild, diffraction-limited blur inherent to AIRR displays. By embedding a differentiable synthetic blur layers into the training loop and employing a two-phase hybrid loss combining MSE, SSIM, LPIPS, and color-consistency terms, the network learns to invert a family of blur kernels in a single model. On test images blurred up to $\sigma = 10$ px, the U-Net improves PSNR by up to 0.30 dB and SSIM by up to 0.002 over the baseline on Gaussian blur, and PSNR improvement up to 1.26 dB and SSIM by 0.0124 on Fraunhofer Diffraction (triangular), while avoiding ringing or color artifacts. We will go on to implement real AIRR PSFs to make the model more robust and reliable further down the pipeline.

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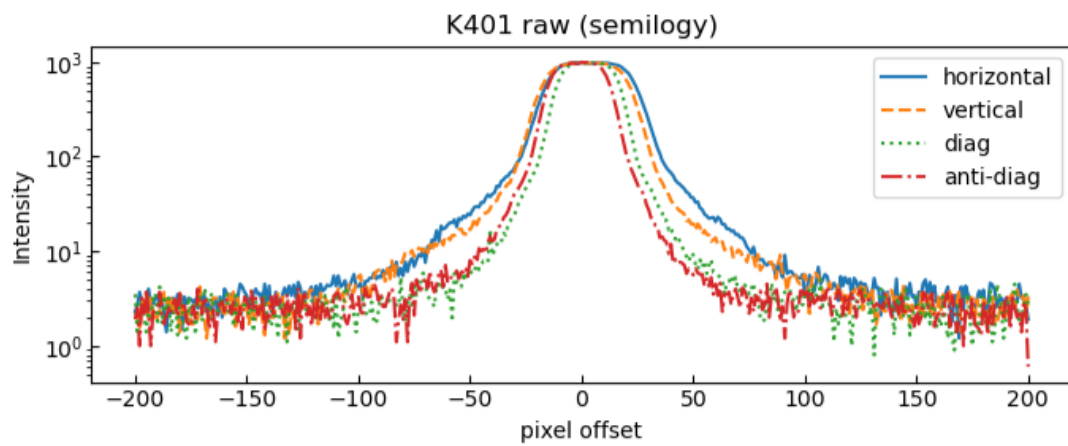


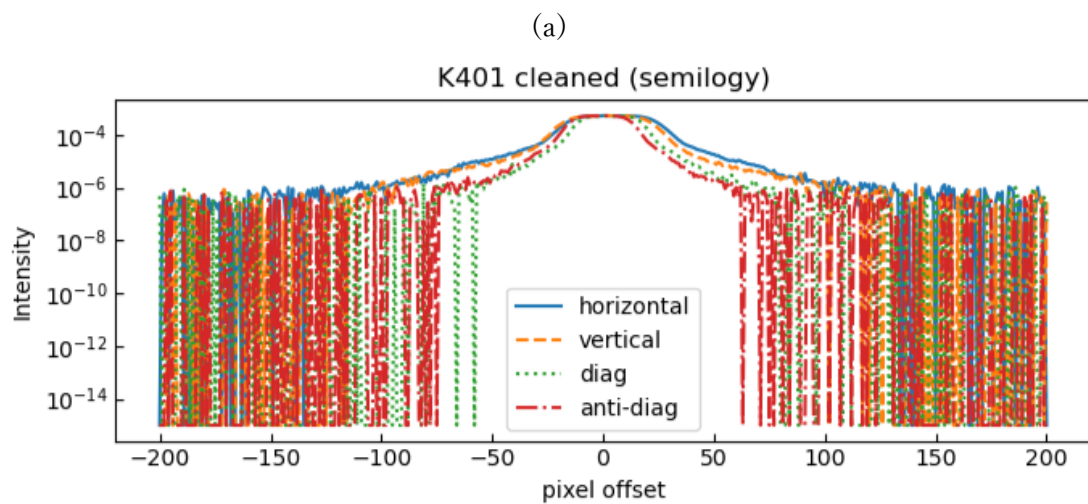
(a)



(b)

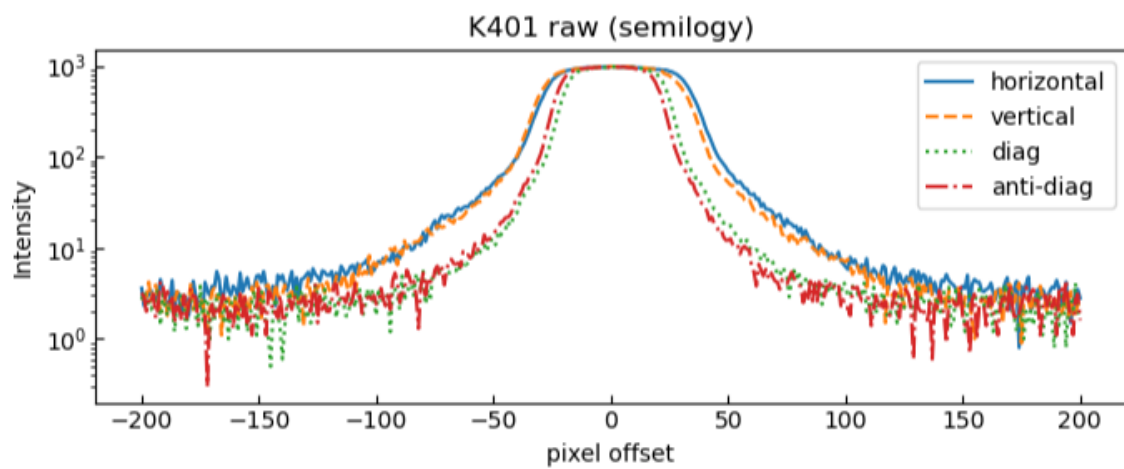
Fig. 30 Logarithmic-scale PSF cross-sections before(a) and after(b) processing
1mm aperture



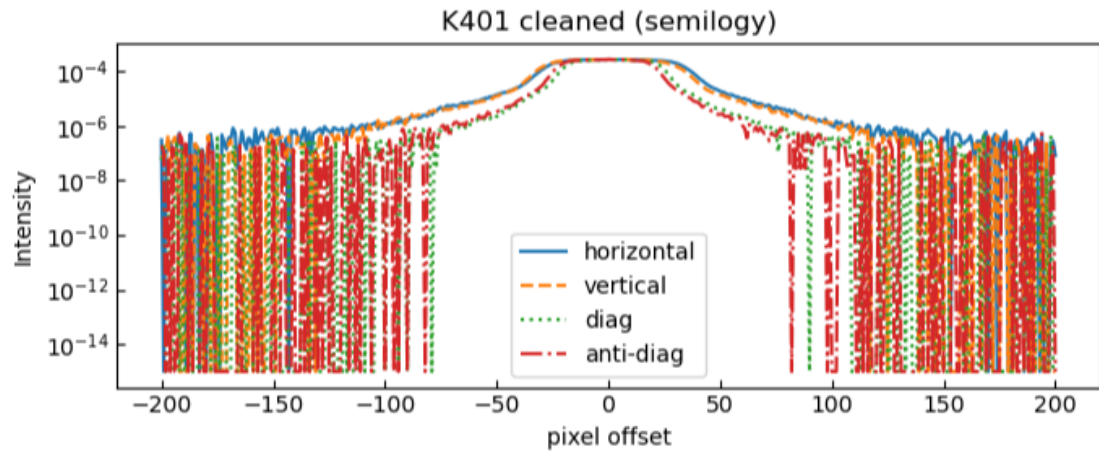


(b)

Fig. 32 Logarithmic-scale PSF cross-sections before(a) and after(b) processing
2mm aperture

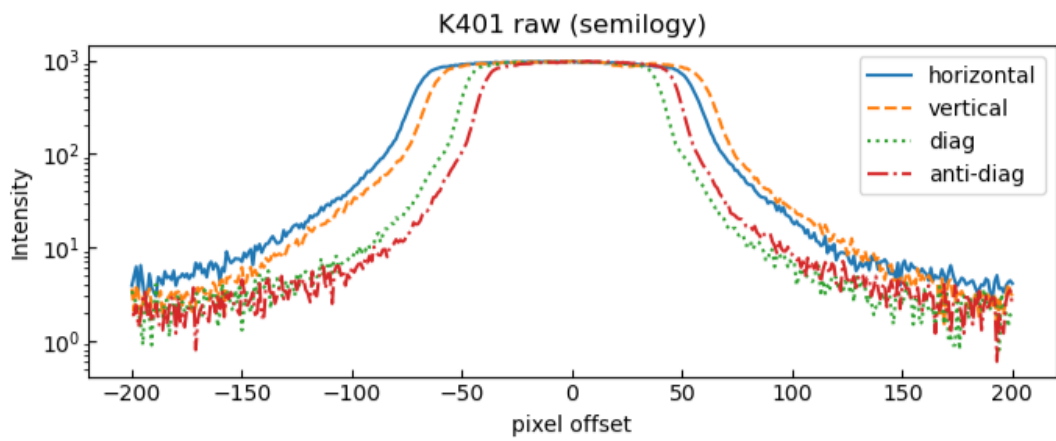


(a)

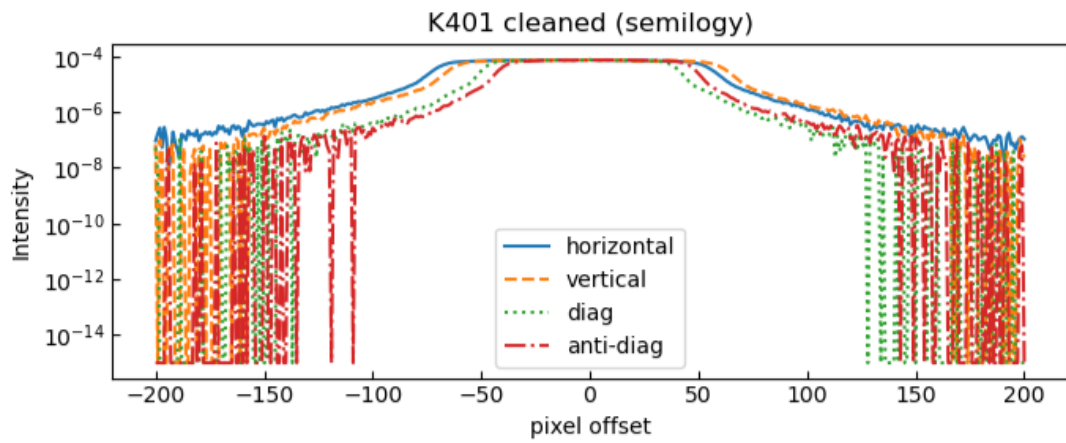


(b)

Fig. 33 Logarithmic-scale PSF cross-sections before(a) and after(b) processing 3mm aperture

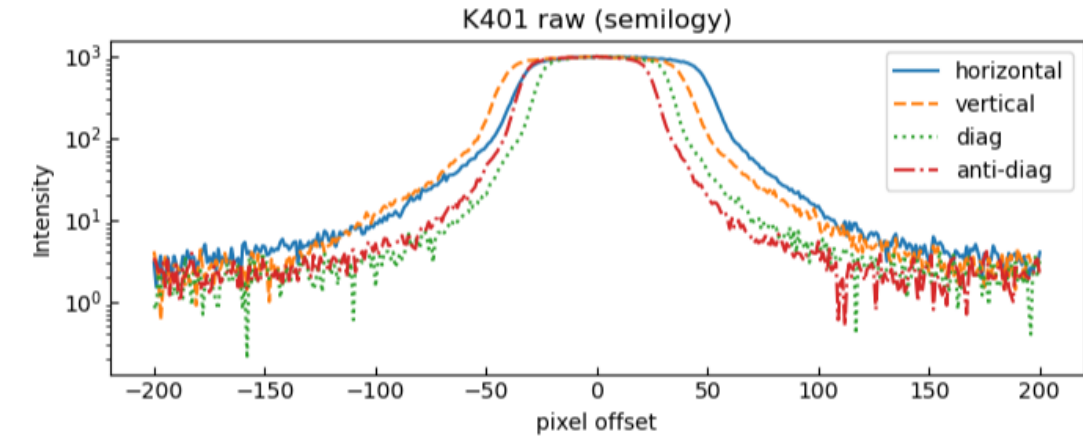


(a)

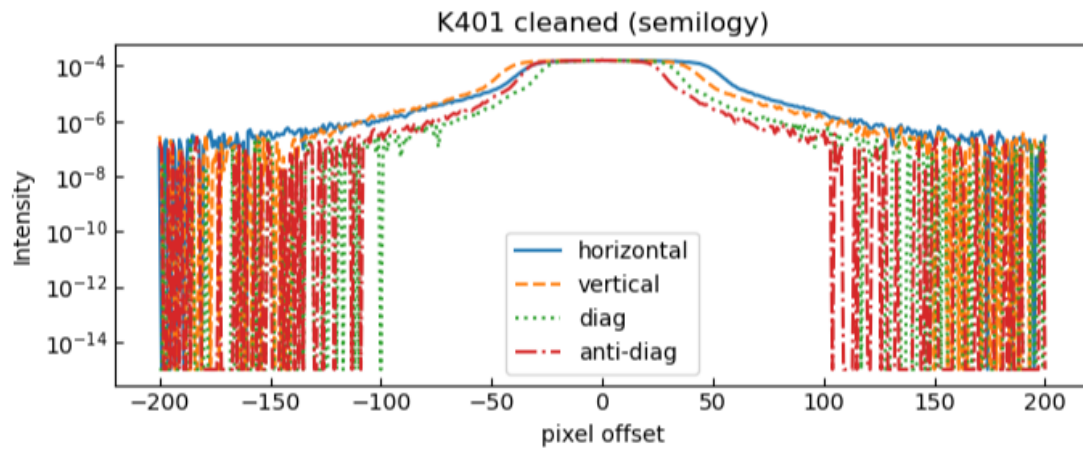


(b)

Fig. 34 Logarithmic-scale PSF cross-sections before(a) and after(b) processing

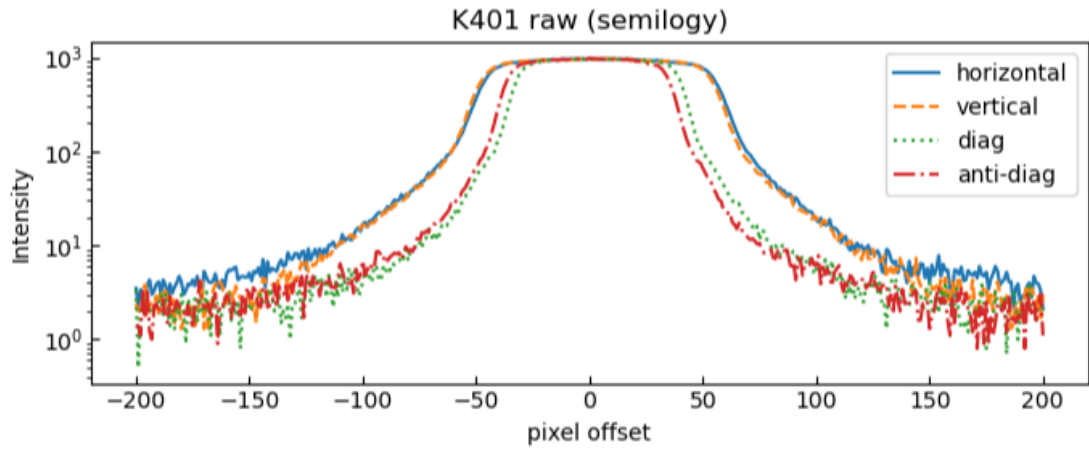


(a)

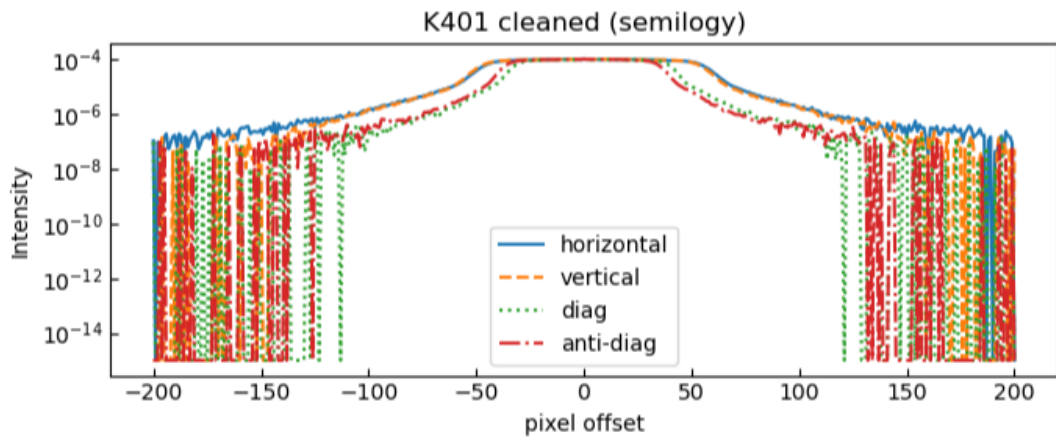


(b)

Fig. 35 Logarithmic-scale PSF cross-sections before(a) and after(b) processing
4mm aperture



(a)



(b)

Fig. 36 Logarithmic-scale PSF cross-sections before(a) and after(b) processing
5mm aperture

List of Abbreviations

AIRR - Aerial Imaging by Retro-Reflection

PSF - Point Spread Function

AI – Artificial Intelligence

CNN – Convolutional Neural Network